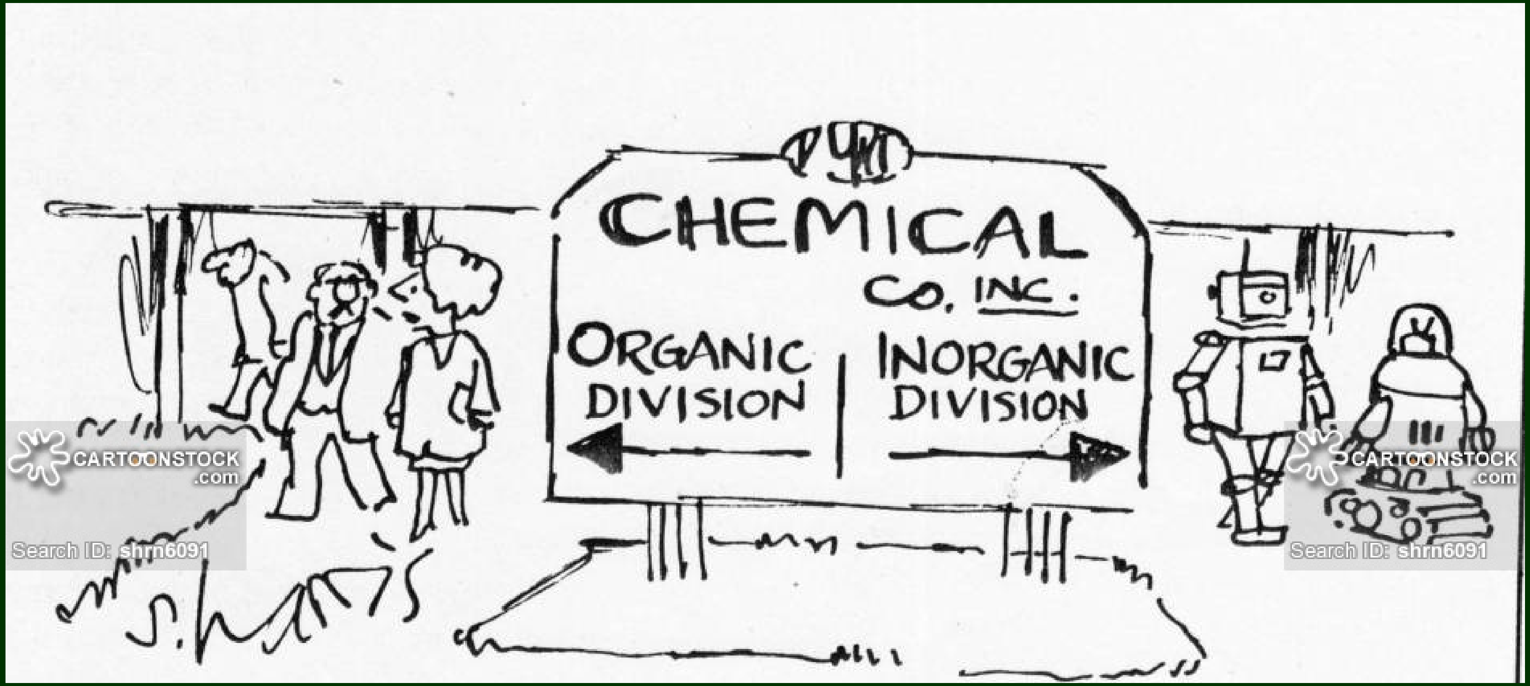
Organic Geochemistry (2)

Organic chemistry for geoscientists
Composition of organic matter



张朝晖

浙江大学舟山校区

2024年09月12日

Structure and Bonding

Molecular Orbital Theory

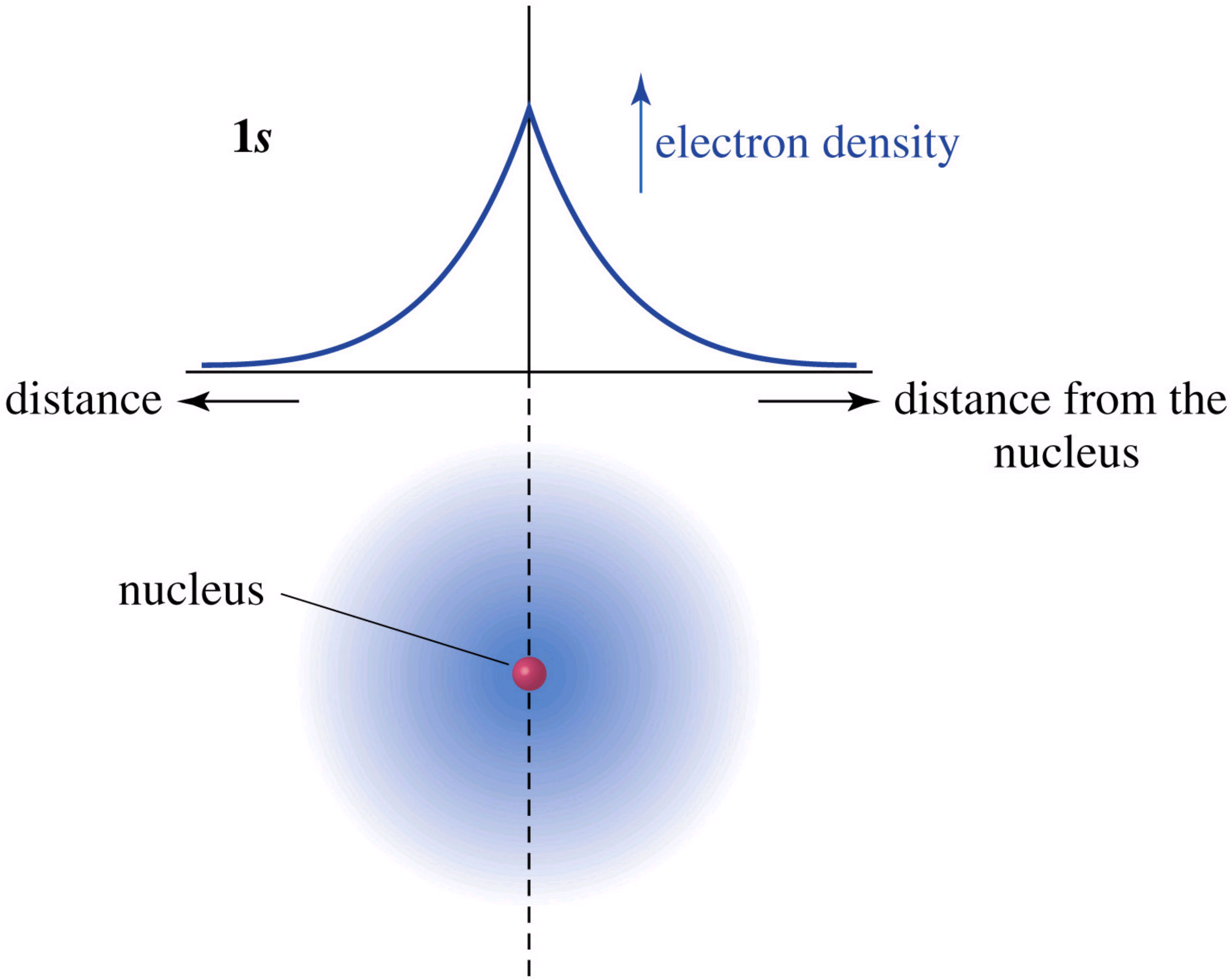
Hybridization

Isomerization

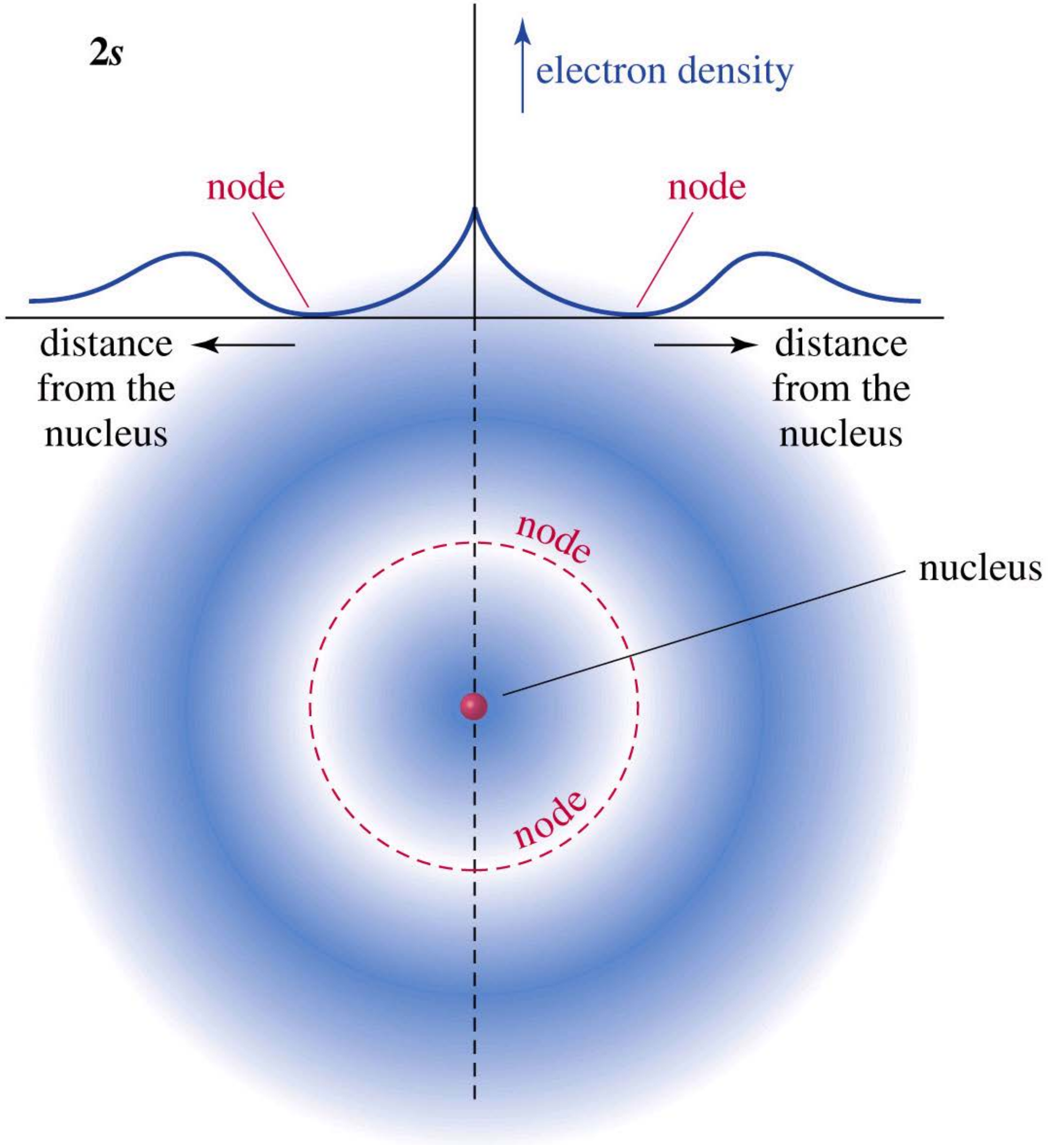
Dipoles and Polarity

Orbitals are Probabilities

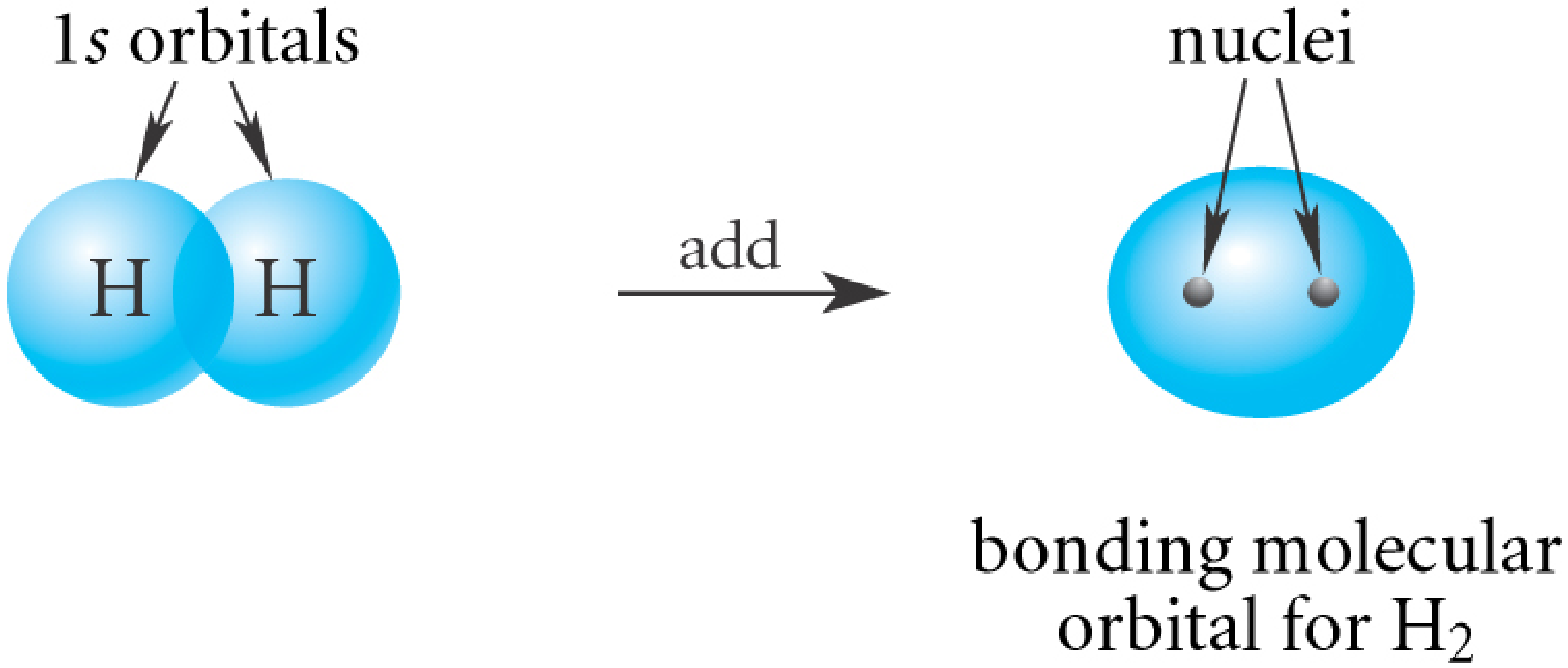
Motion of Electrons Described as Wave Functions Ψ



2s Orbital Has a Node



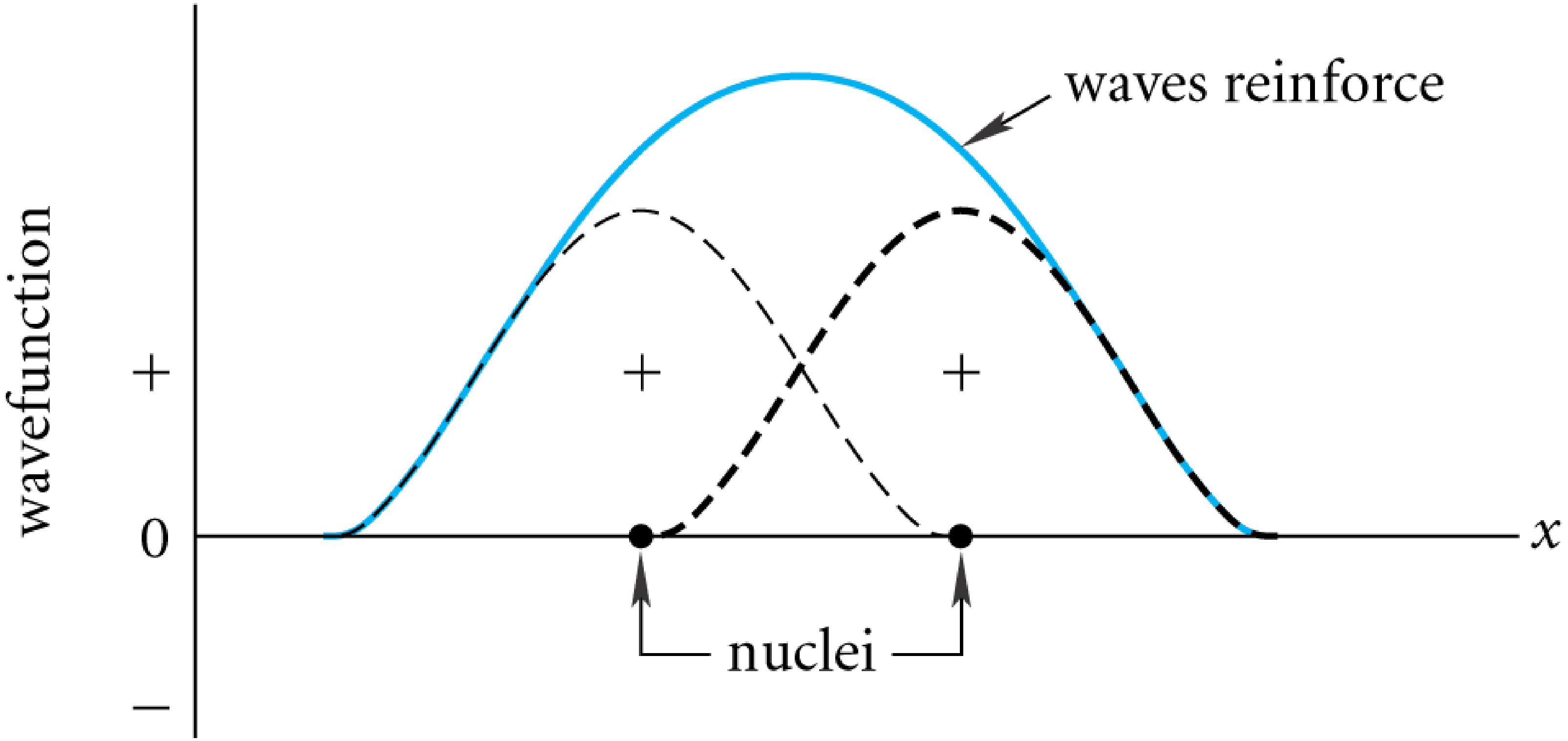
Bonding in H_2
Sigma (s) Bond



(b)

Wave Depiction of H_2

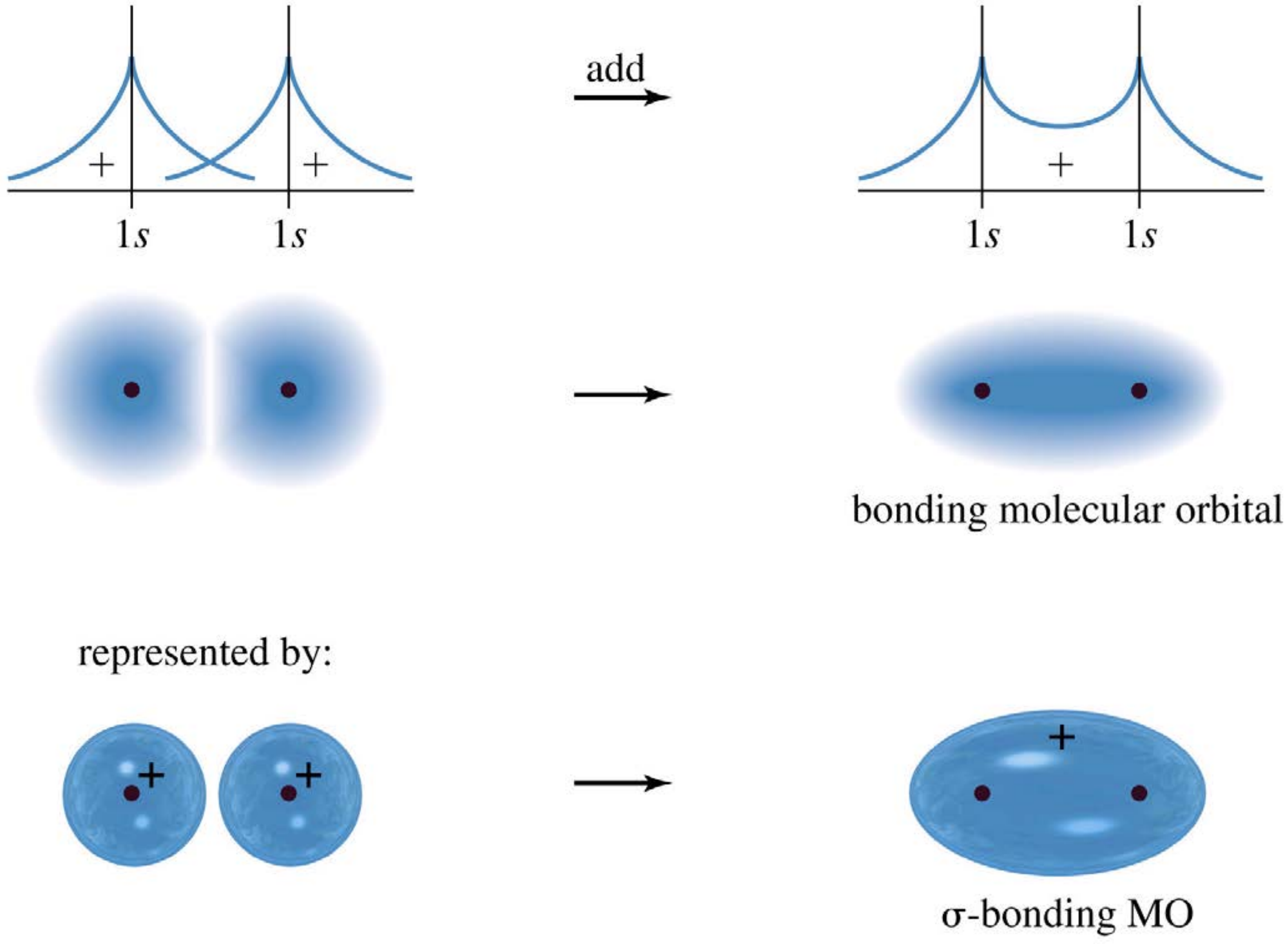
Waves reinforce



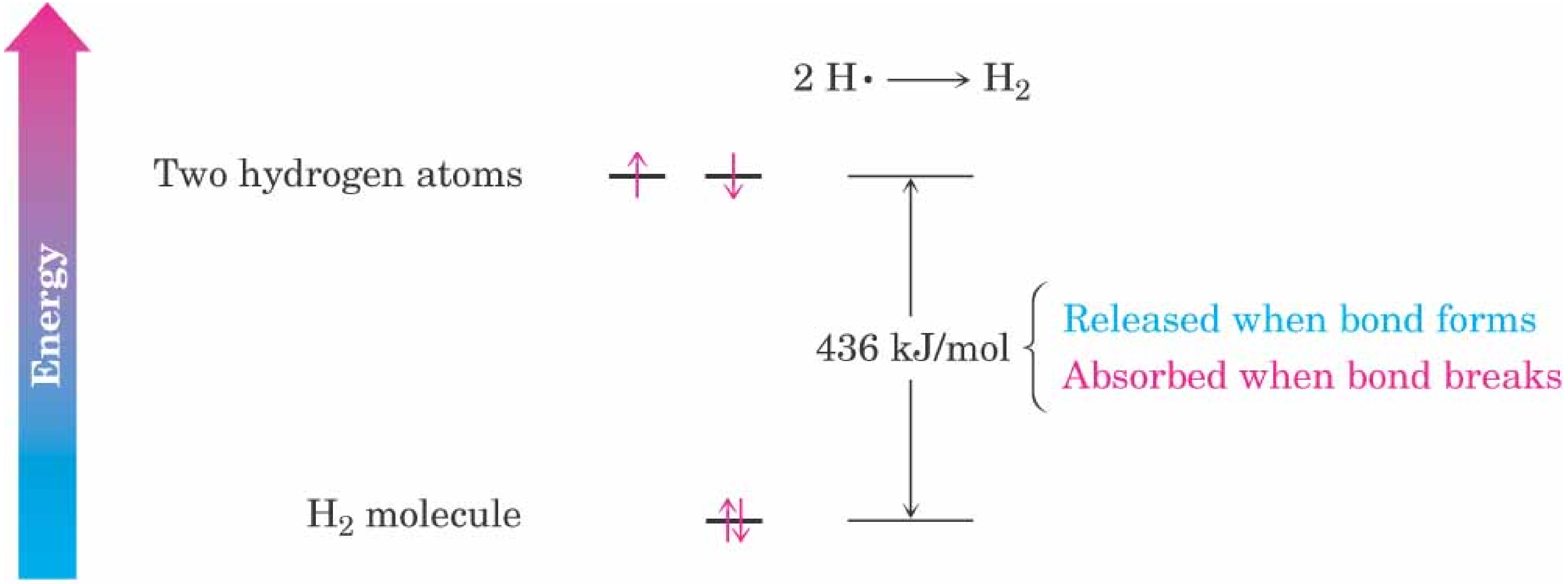
(a)

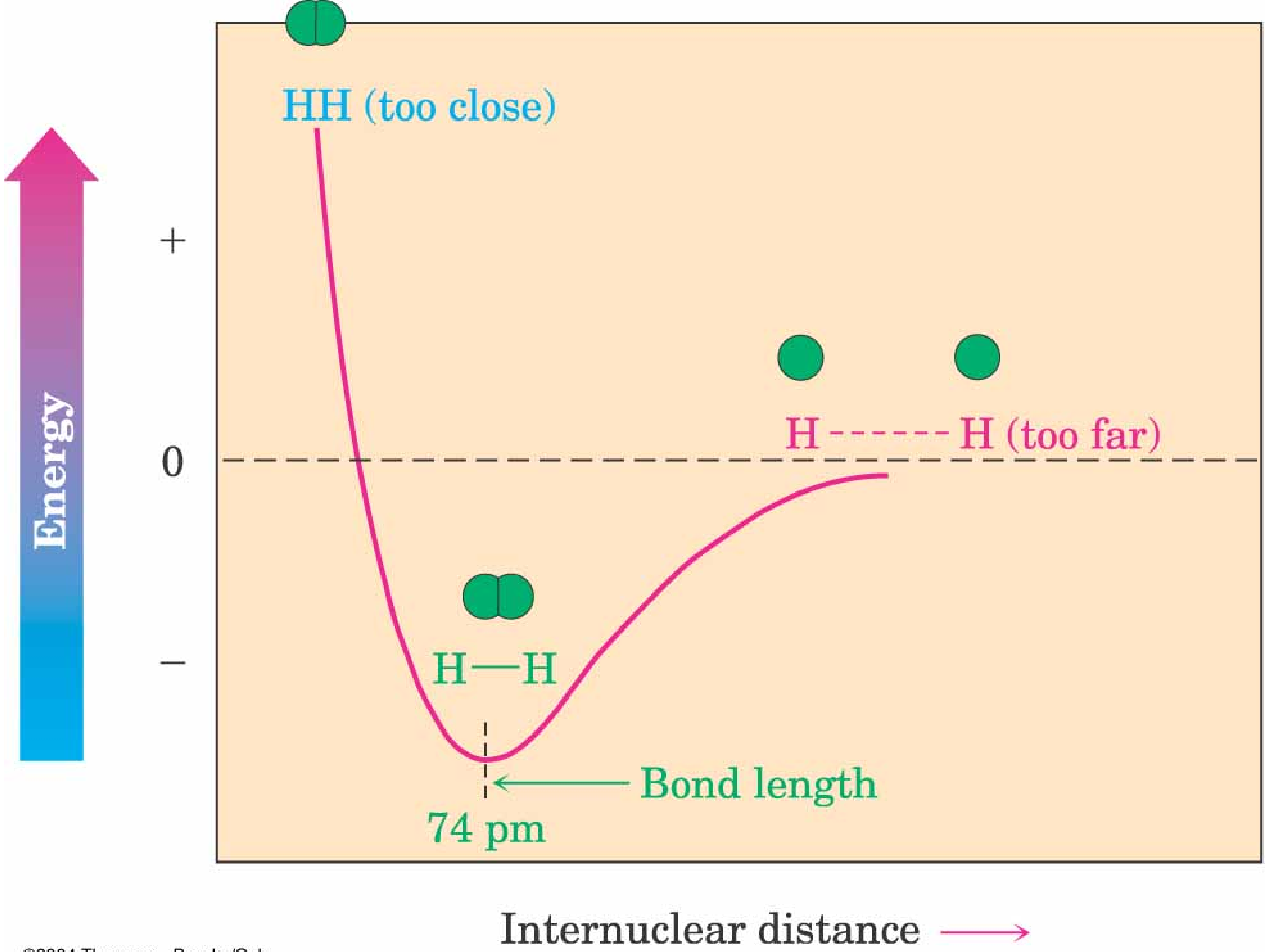
Molecular Orbitals

Mathematical Combination of Atomic Orbitals



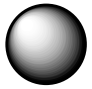
Energy Released When Bond Forms



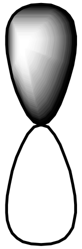


HH (too close)
Energy
+
0
−
H - - - - - - H (too far)
H—H
Bond length
74 pm
Internuclear distance

Electronic configuration of hydrogen H $1s^1$

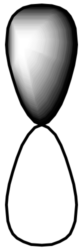


H 1s
(1 $e^{\ominus}$)



C $2p_y$

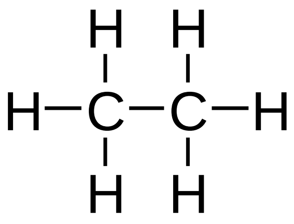
(vacant)



C $2p_z$

(vacant)

Electronic configuration of Carbon C $1s^2\ 2s^2\ 2p^2$



Ethane

Orbitals available for covalent bonding?

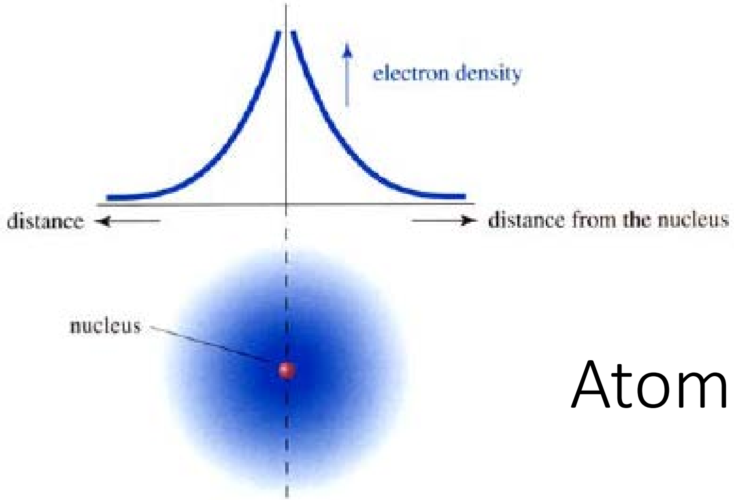
Ground State Electron Configurations

TABLE 1.1 Ground-State Electron Configurations of Some Elements

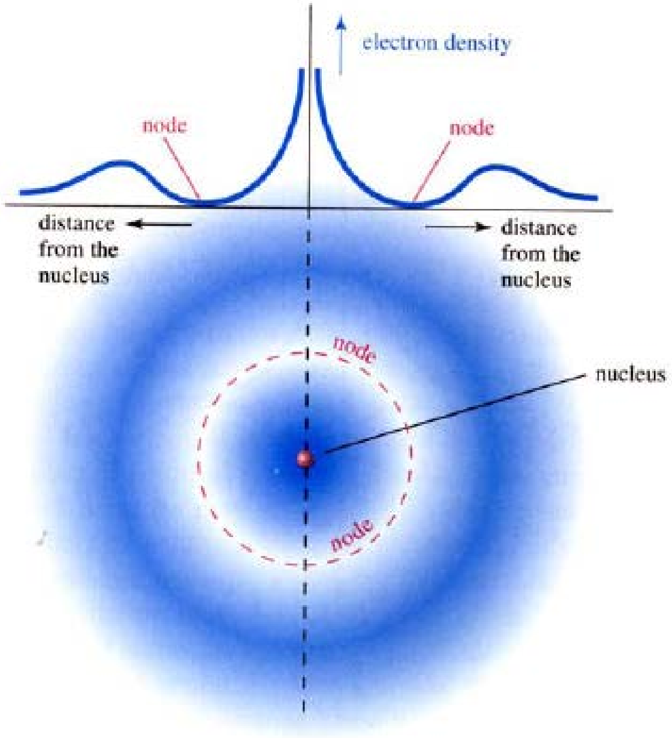
Element	Atomic number	Configuration	Element	Atomic number	Configuration
Hydrogen	1	1*s* ↑	Lithium	3	2*s* ↑ 1*s* ↑↓
Carbon	6	2*p* ↑ ↑ — 2*s* ↑↓ 1*s* ↑↓	Neon	10	2*p* ↑↓ ↑↓ ↑↓ 2*s* ↑↓ 1*s* ↑↓
Sodium	11	3*s* ↑ 2*p* ↑↓ ↑↓ ↑↓ 2*s* ↑↓ 1*s* ↑↓	Argon	18	3*p* ↑↓ ↑↓ ↑↓ 3*s* ↑↓ 2*p* ↑↓ ↑↓ ↑↓ 2*s* ↑↓ 1*s* ↑↓

Atomic electron orbitals

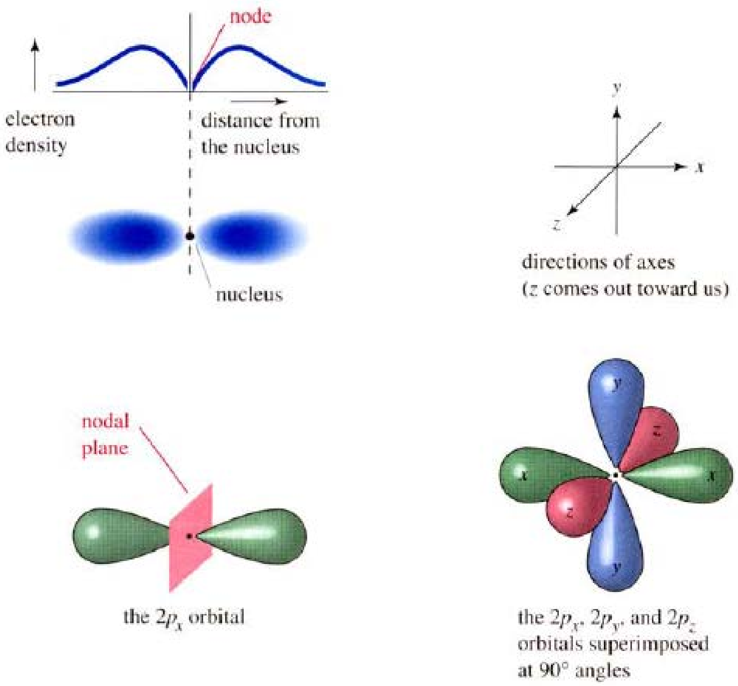
1S



1P

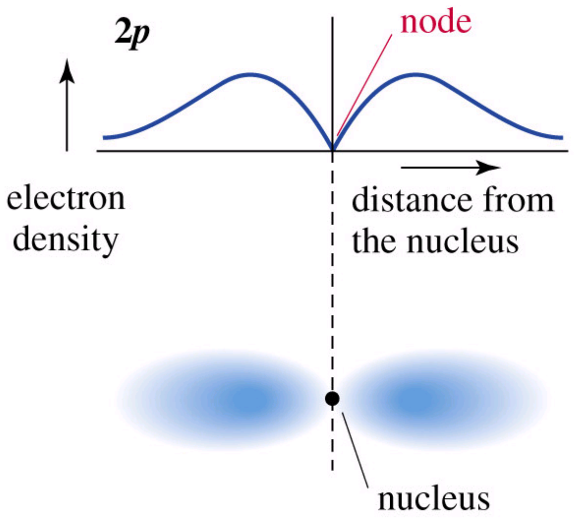


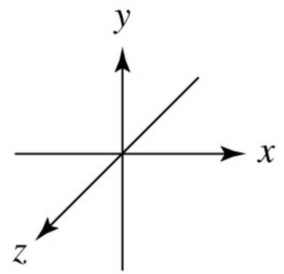
2P



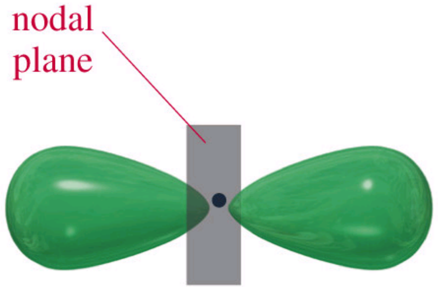
From L. Wade Organic Chemistry

The p Orbital

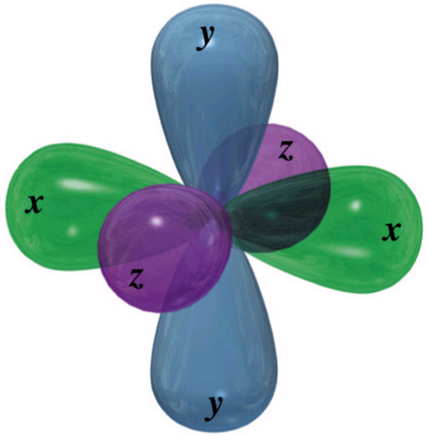




directions of axes
(z comes out toward us)

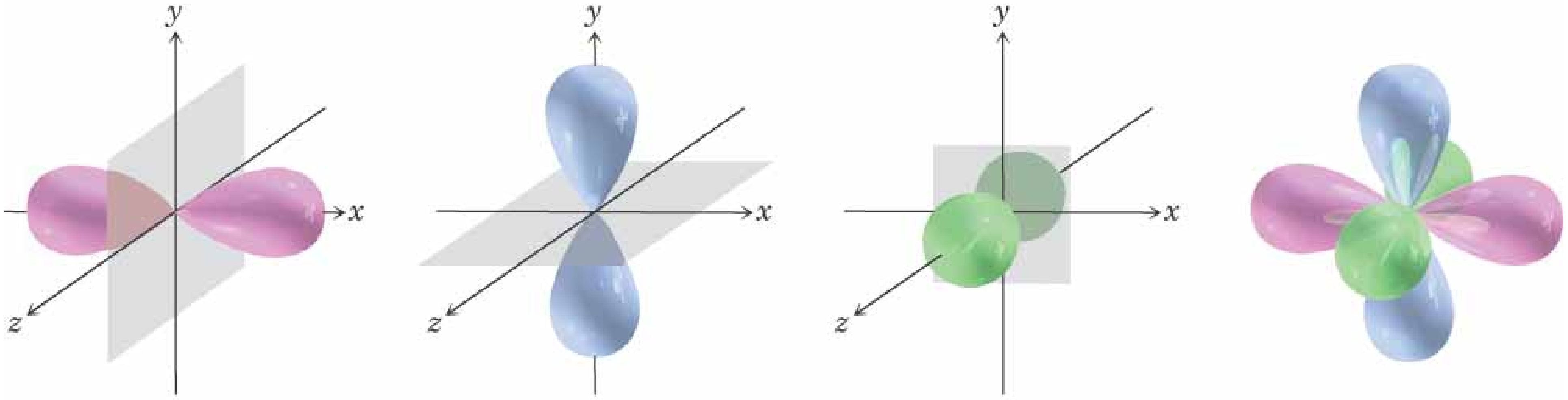


the $2p_x$ orbital



the $2p_x$, $2p_y$, and $2p_z$
orbitals superimposed
at 90° angles

p_x, p_y, p_z

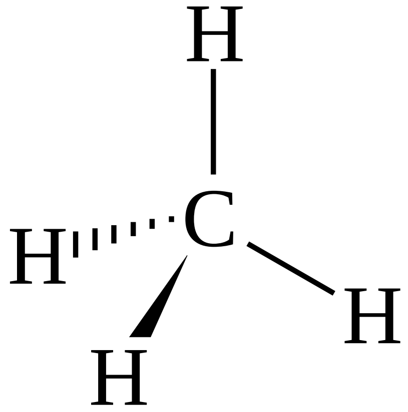


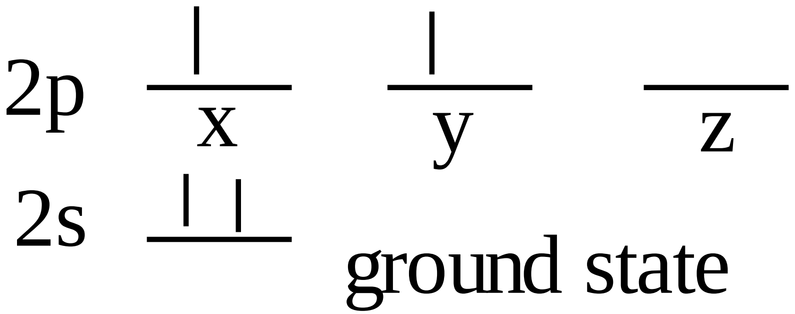
A $2p_x$ orbital | A $2p_y$ orbital | A $2p_z$ orbital | Three $2p$ orbitals

In Ground State

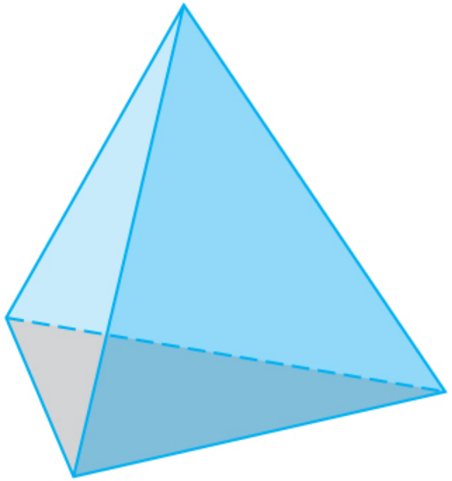
2 bonding sites, 1 lone pair

CH_4

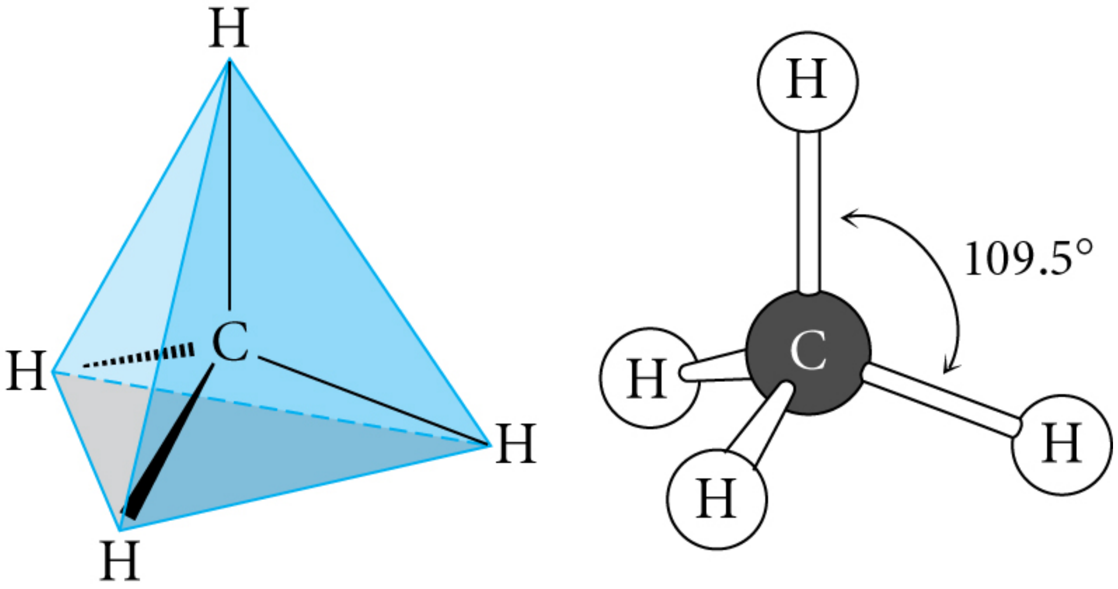




Tetrahderal Geometry



a regular tetrahedron

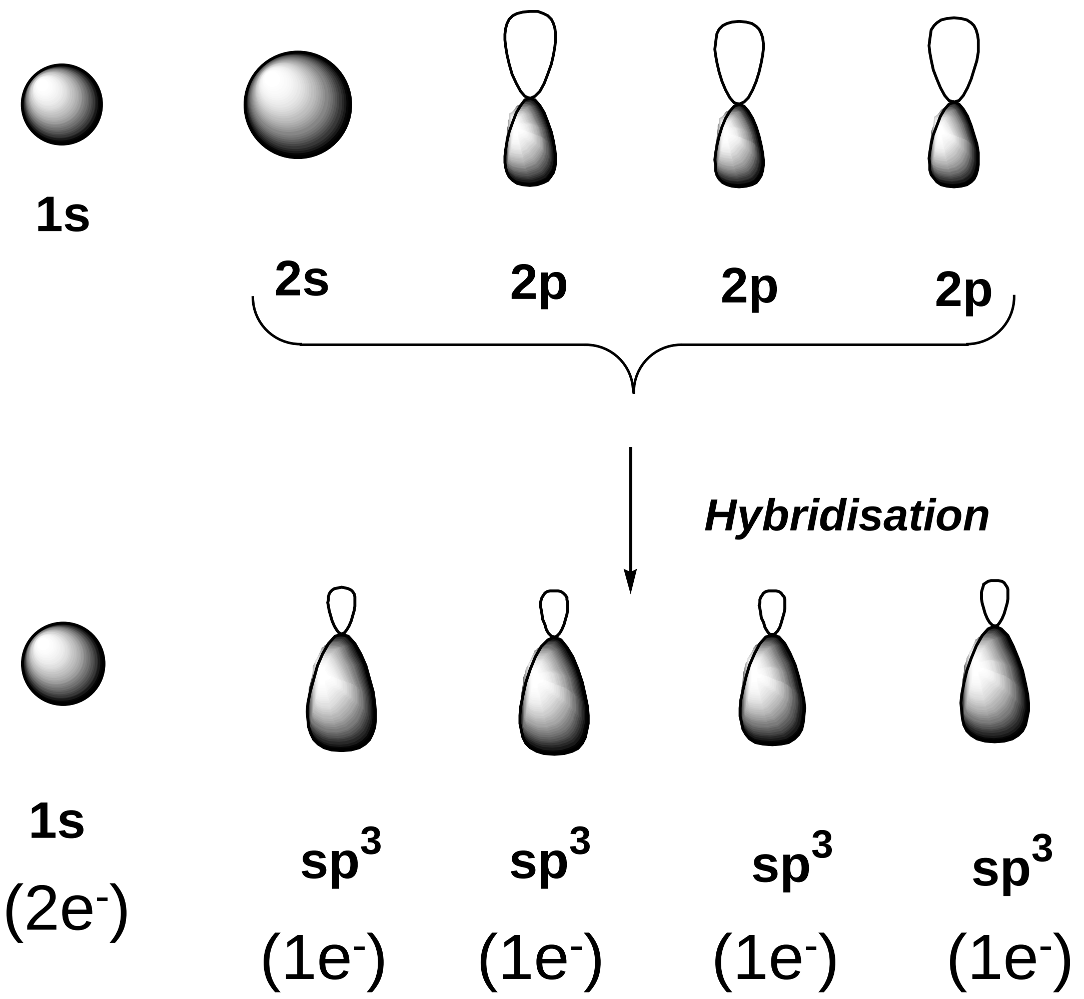


methane

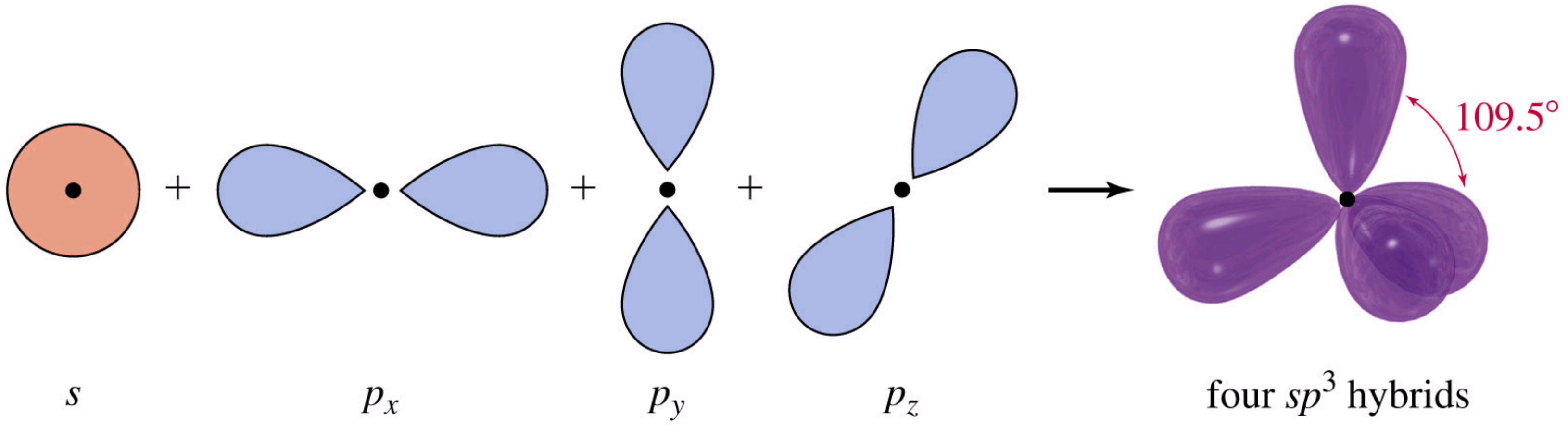
•However, know that the geometry of the Carbons in ethane is tetrahedral

•Cannot array p_y and p_z orbitals to give tetrahedral geometry

•Need a modified set of atomic orbitals - *hybridization*



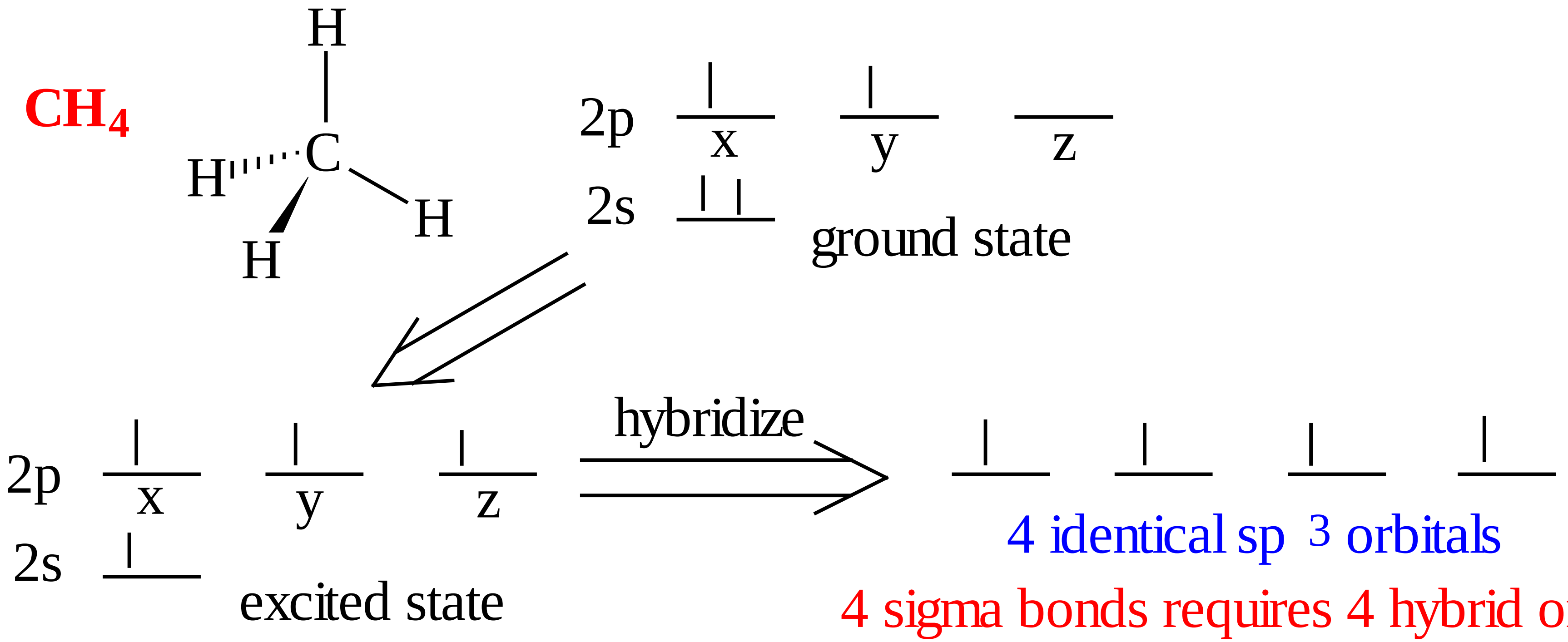
Hybridization of 1 s and 3 p Orbitals – sp^3



sp^3 Hybridization

4 Regions of electron Density

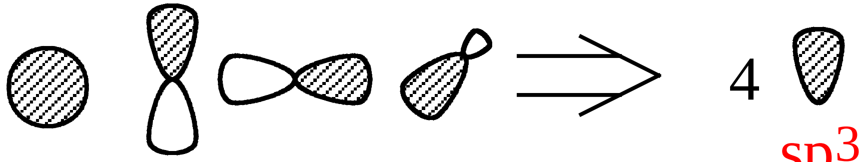
CH_4



4 identical sp^3 orbitals

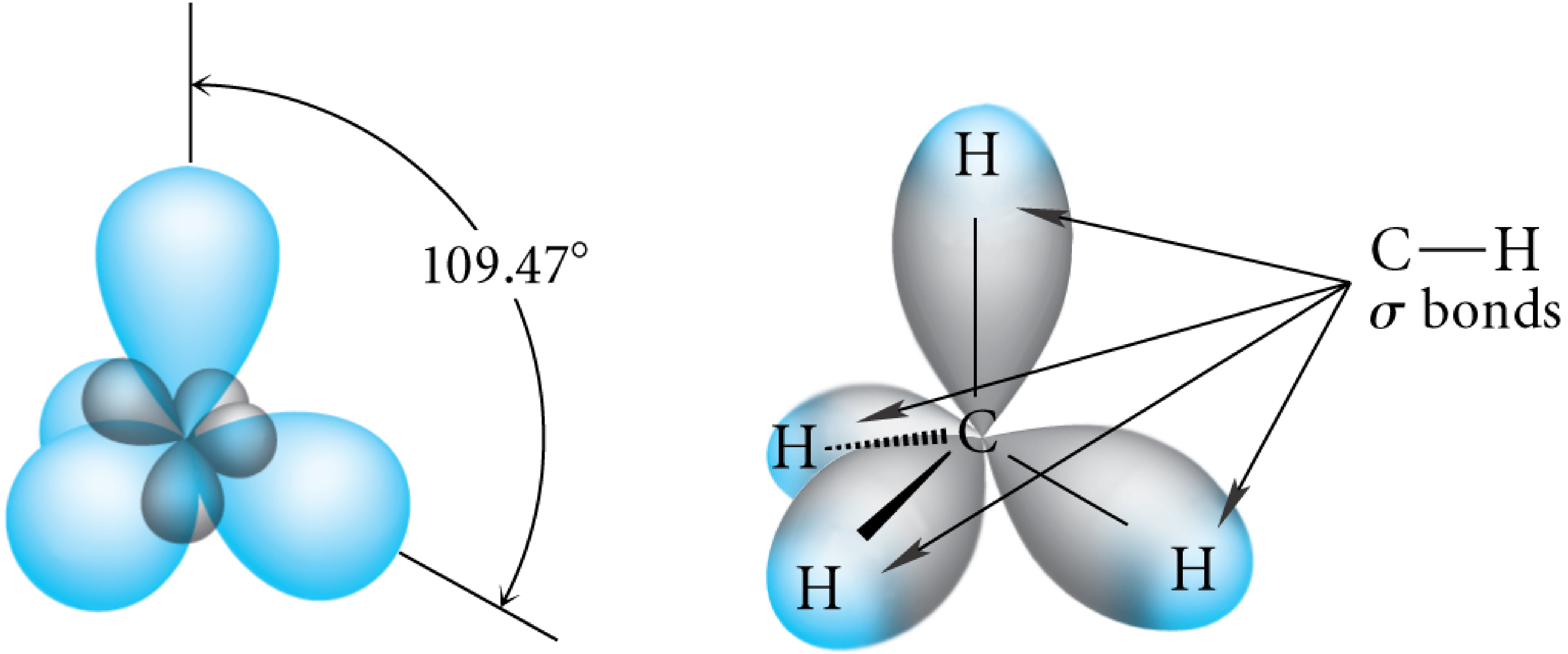
4 sigma bonds requires 4 hybrid orbitals

tetrahedral geometry



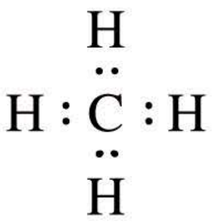
sp^3

sp^3 is Tetrahedral Geometry
Methane

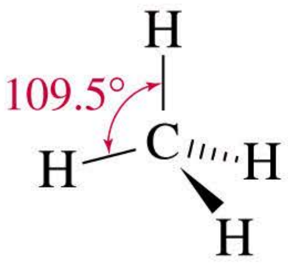


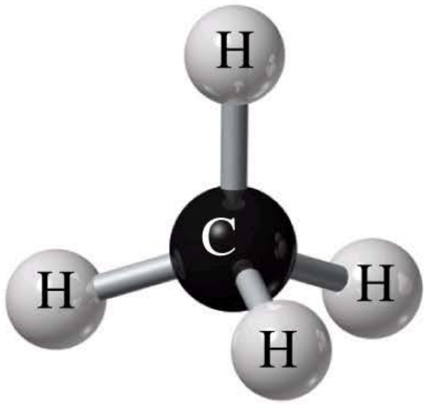
(c) (d)

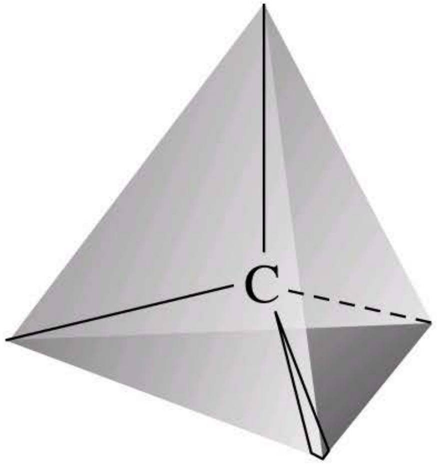
Methane Representations

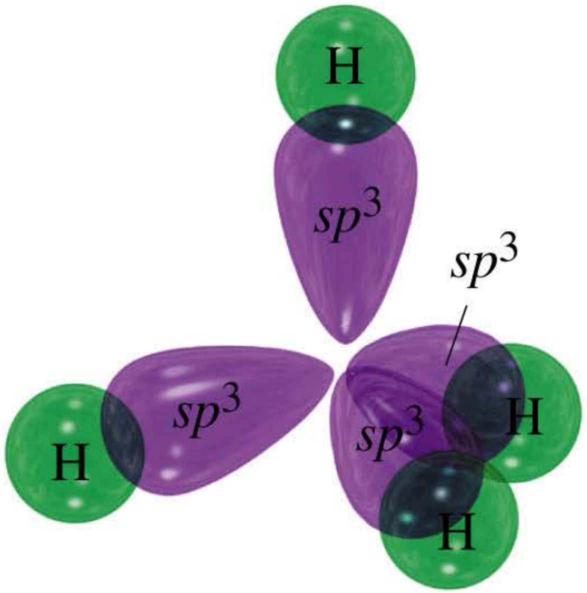


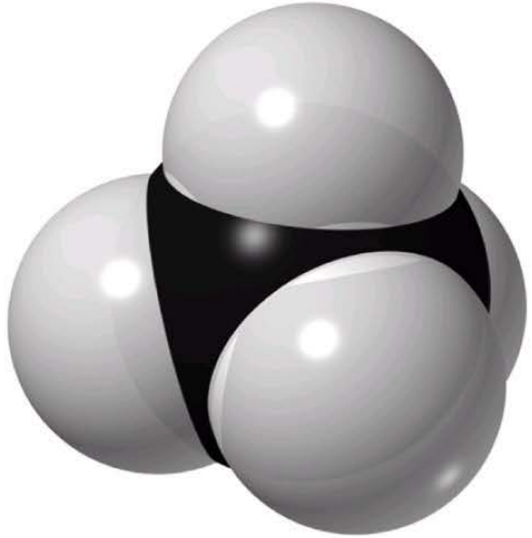
methane





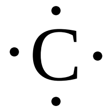






Lewis Dot Structure of Methane

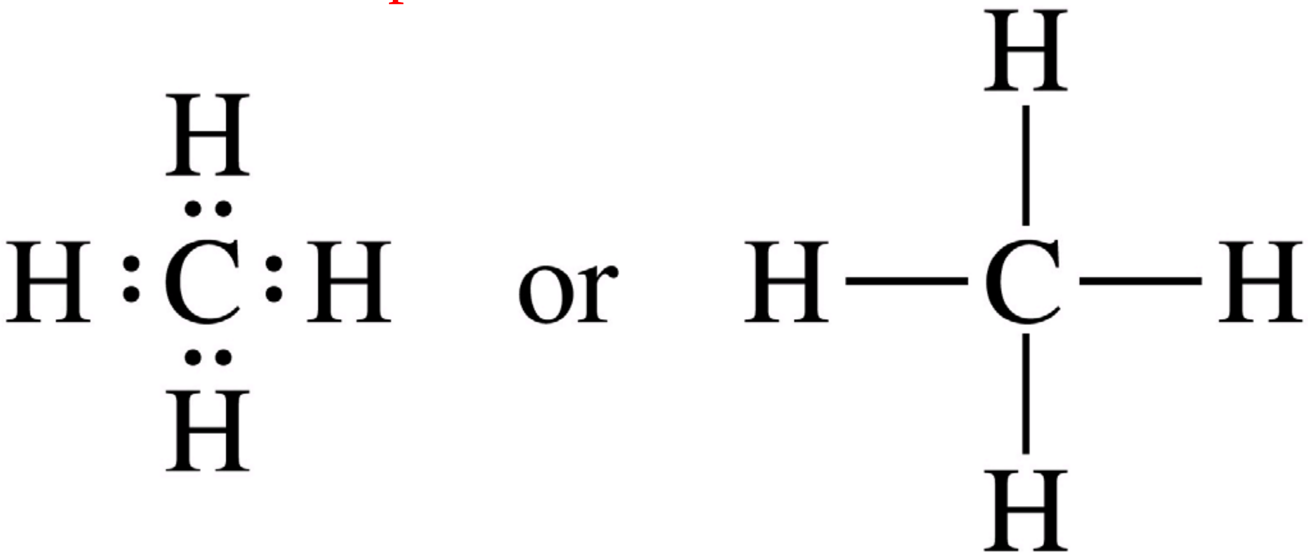
carbon - 4 valence e



$1s^2$$2s^22p^2$

hydrogen - 1 valence e

1s

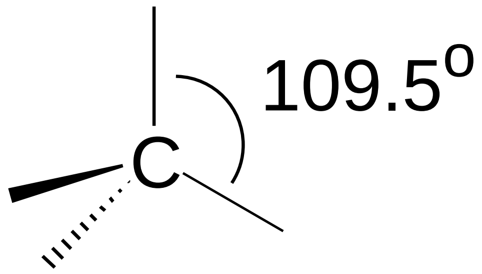


methane

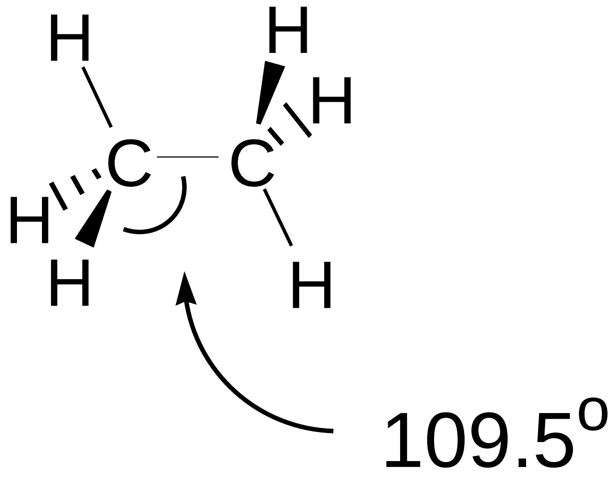
Geometry of Carbon in ethane is tetrahedral and is based upon sp^3 hybridisation

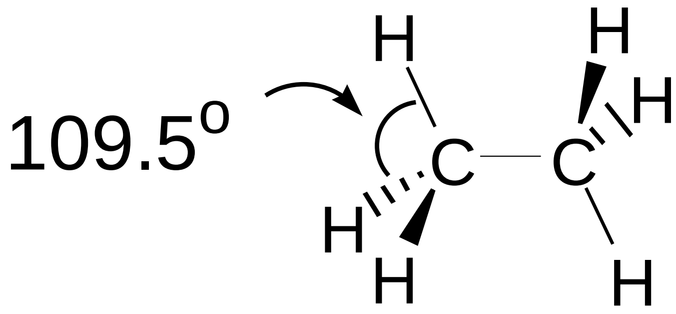
sp^3 hybridised Carbon = tetrahedral Carbon

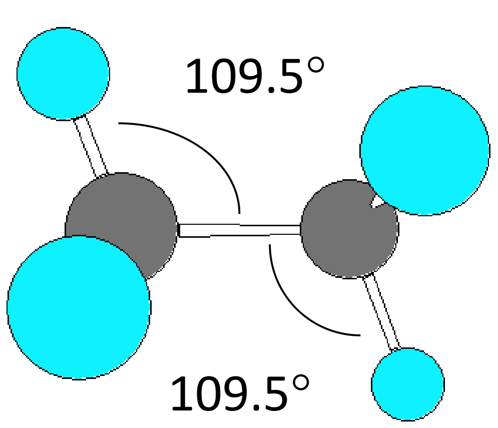
Tetrahedral angle ≈ 109.5^o

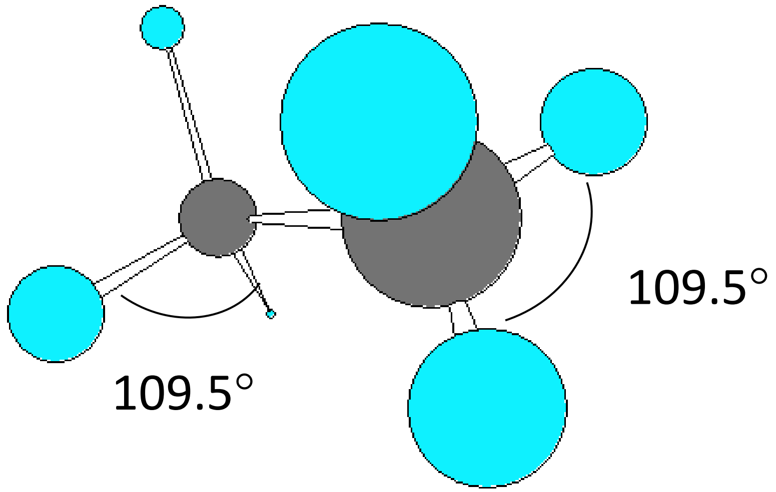


Angle between any two bonds at a Carbon atom = 109.5°

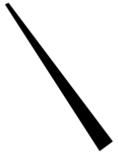


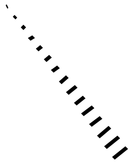






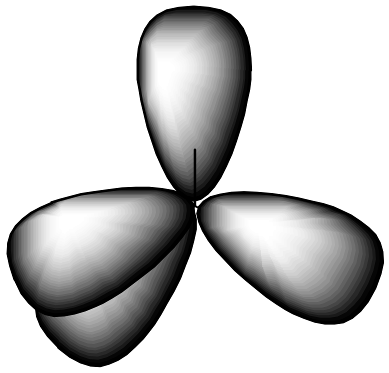
Need to be able to represent 3D molecular structure in 2D

 $\Longrightarrow$ Bond coming out of plane of screen

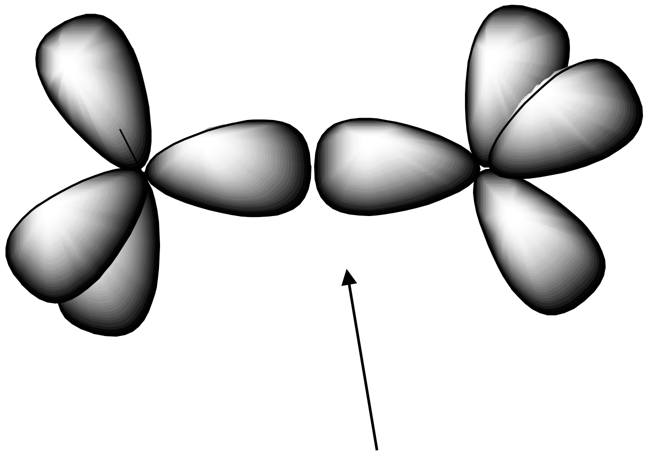
 $\Longrightarrow$ Bond going into plane of screen

Visualising the molecular orbitals in ethane

Four sp^3 hybridised orbitals can be arrayed to give tetrahedral geometry



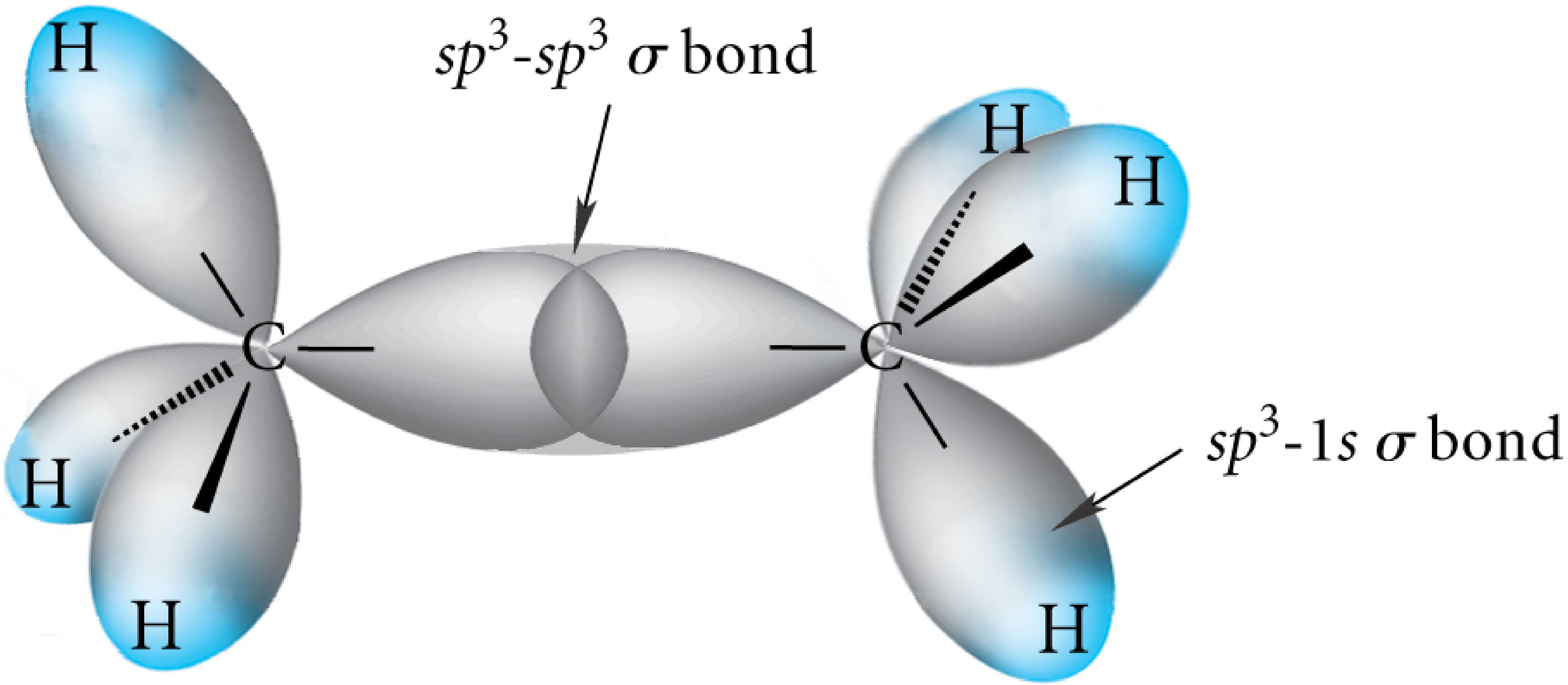
sp^3 hybridised orbitals from two Carbon atoms can overlap to form a Carbon-Carbon σ bond



C-C σ bond

Each sp^3 orbital contributes one electron to form C-C [C··C]

Orbital Depiction of Ethane, C_2H_6 , the σ bond



Electronic configuration of Carbon $C \ 1s^2 \ 2s^2 \ 2p^2$

- *Covalent bonds*: sharing of electrons between atoms
- Carbon: can accept 4 electrons from other atoms
- i.e. **Carbon is tetravalent (valency = 4)**

Ethane: a gas (b.p. ~ -88.6°C)

Empirical formula (elemental combustion analysis): CH_3

i.e. an organic chemical

Measure molecular weight (e.g. by mass spectrometry):

30.070 g mol^{-1}, i.e $(CH_3)_n$ $n = 2$

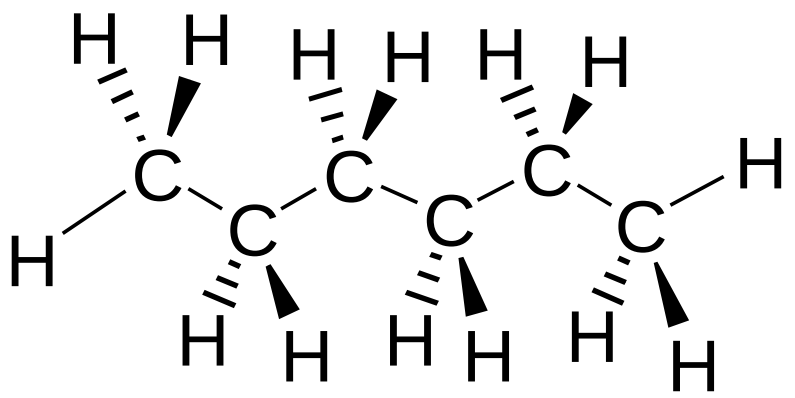
Implies **molecular formula** = C_2H_6

Simple alkanes have conformational freedom at room temperature

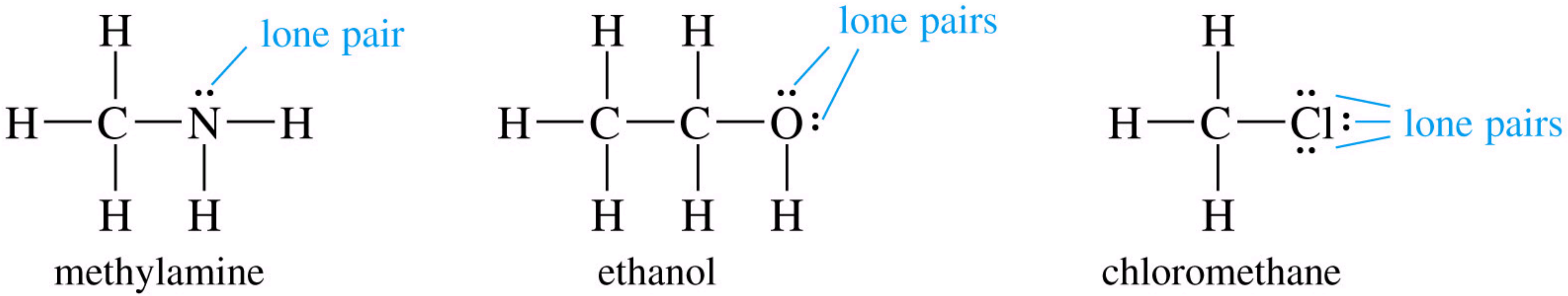
i.e. have rotation about C-C bonds

the most stable (lowest energy) conformation for these is the all staggered 'straight chain'

e.g. for hexane



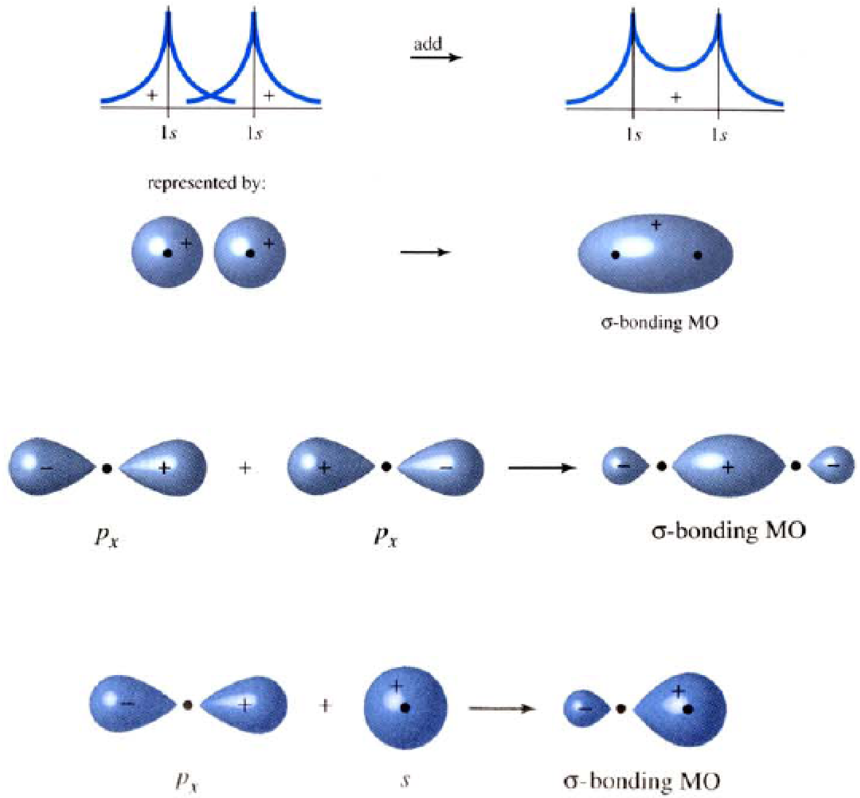
	carbon	nitrogen	oxygen	hydrogen	halogens
valence:	4	3	2	1	1
lone pairs:	0	1	2	0	3



Sigma and pi bonds

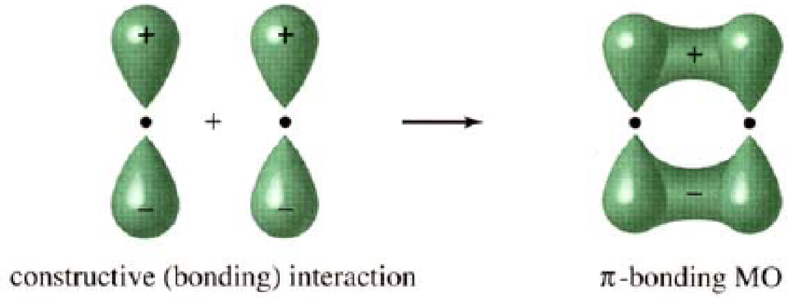
Formation of sigma (σ) bonds.

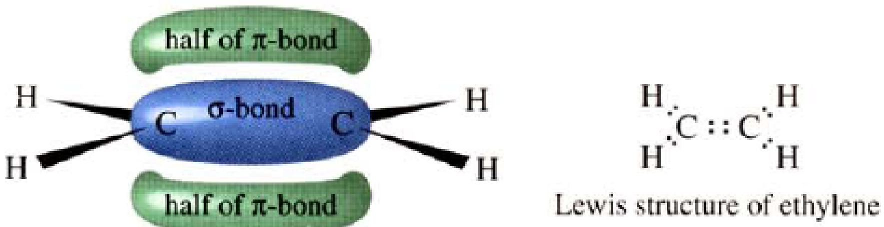
Electron density is highest between nucleii,
and is cylindrically symmetric.



Formation of pi (π) bonds.

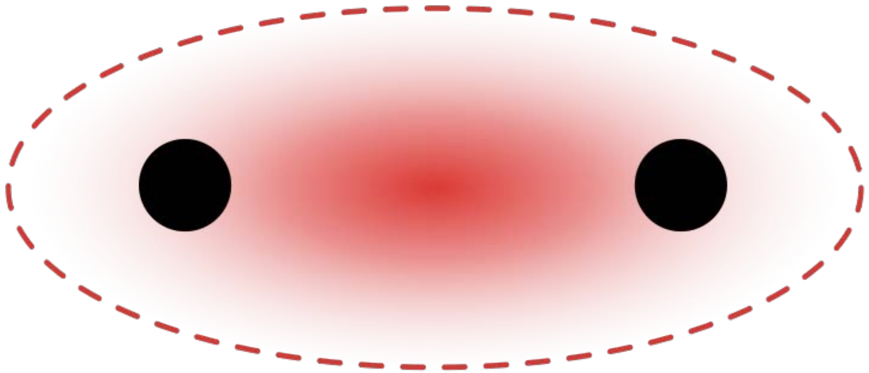
Bonding by parallel overlap of p orbitals.
Electron density is above and below sigma bond.

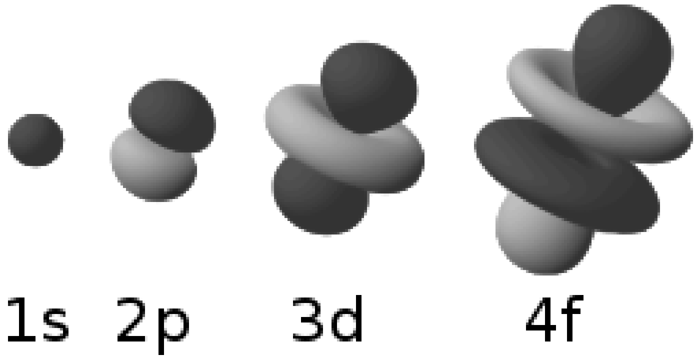


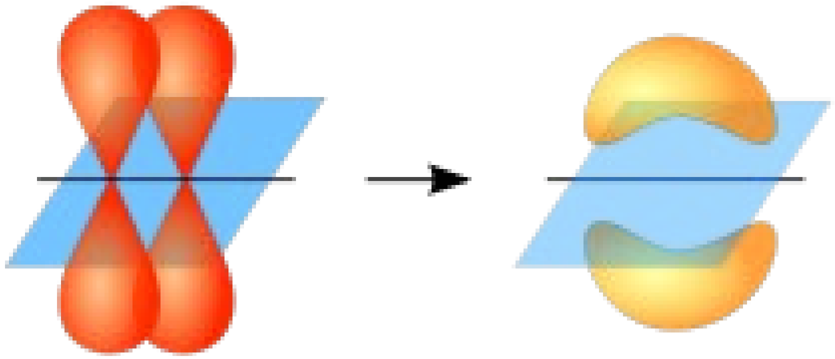


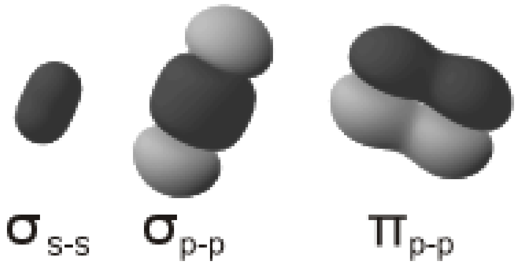
Wade, Organic Chemistry

σ bond: the strongest type of covalent chemical bond.
They are formed by head-on overlapping between atomic orbitals.



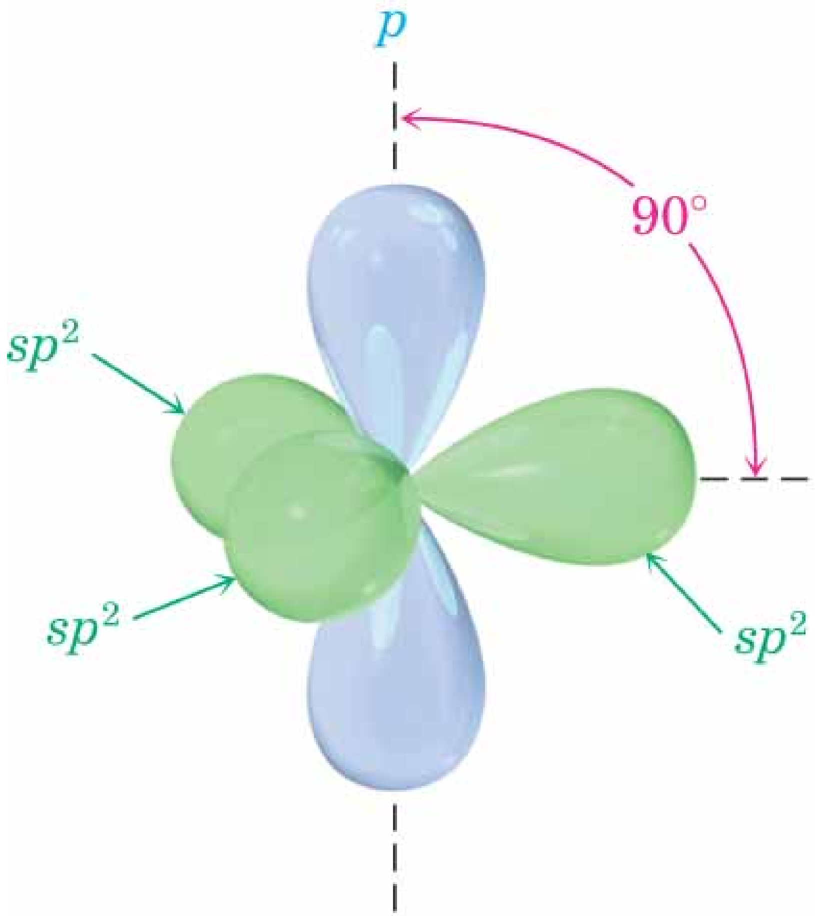




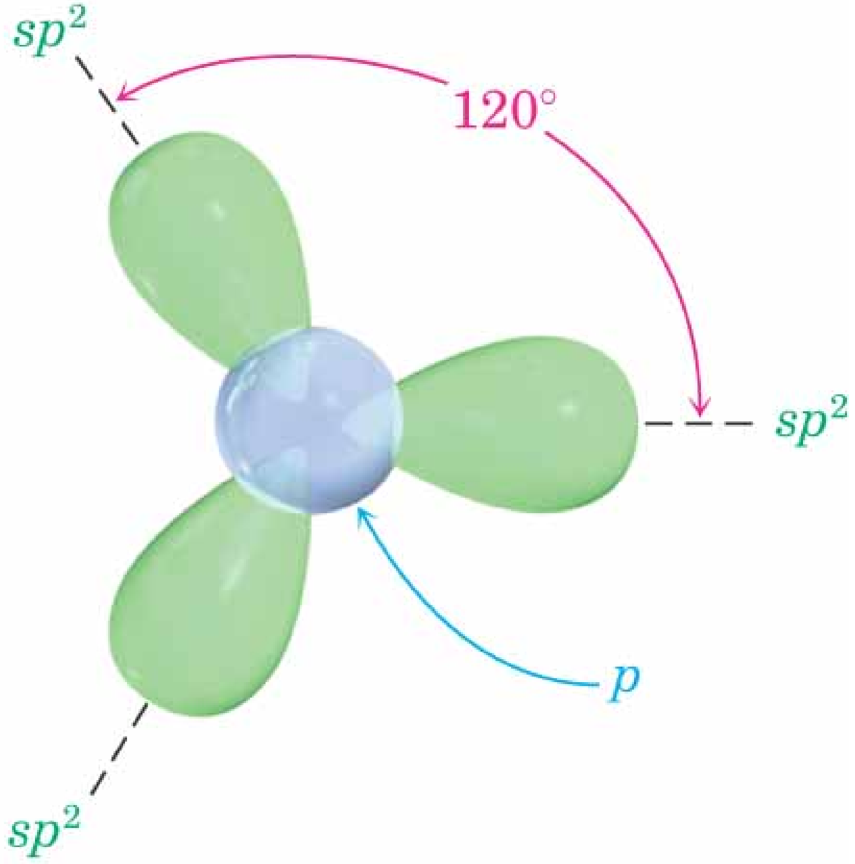


π bond: covalent chemical bonds where two lobes of one involved atomic orbital overlap two lobes of the other involved atomic orbital.

An sp^2 Hybridized Atom

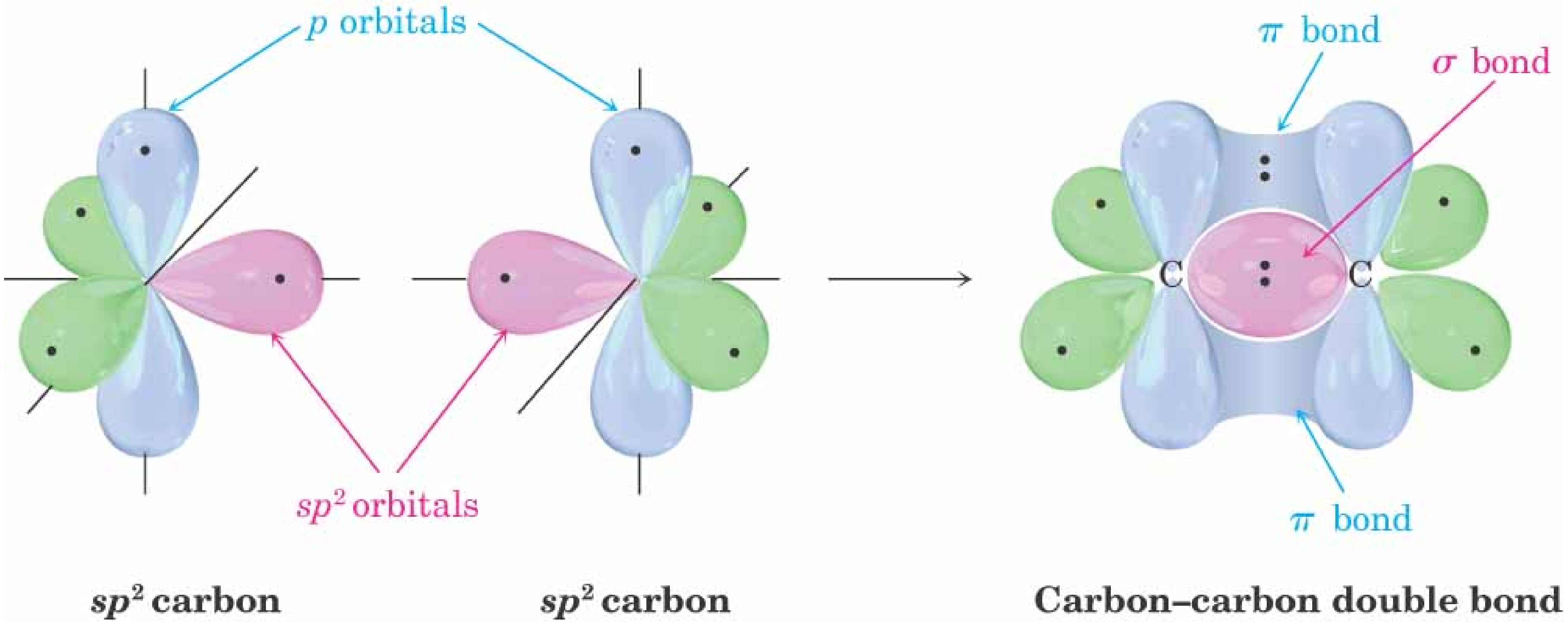


Side view



Top view

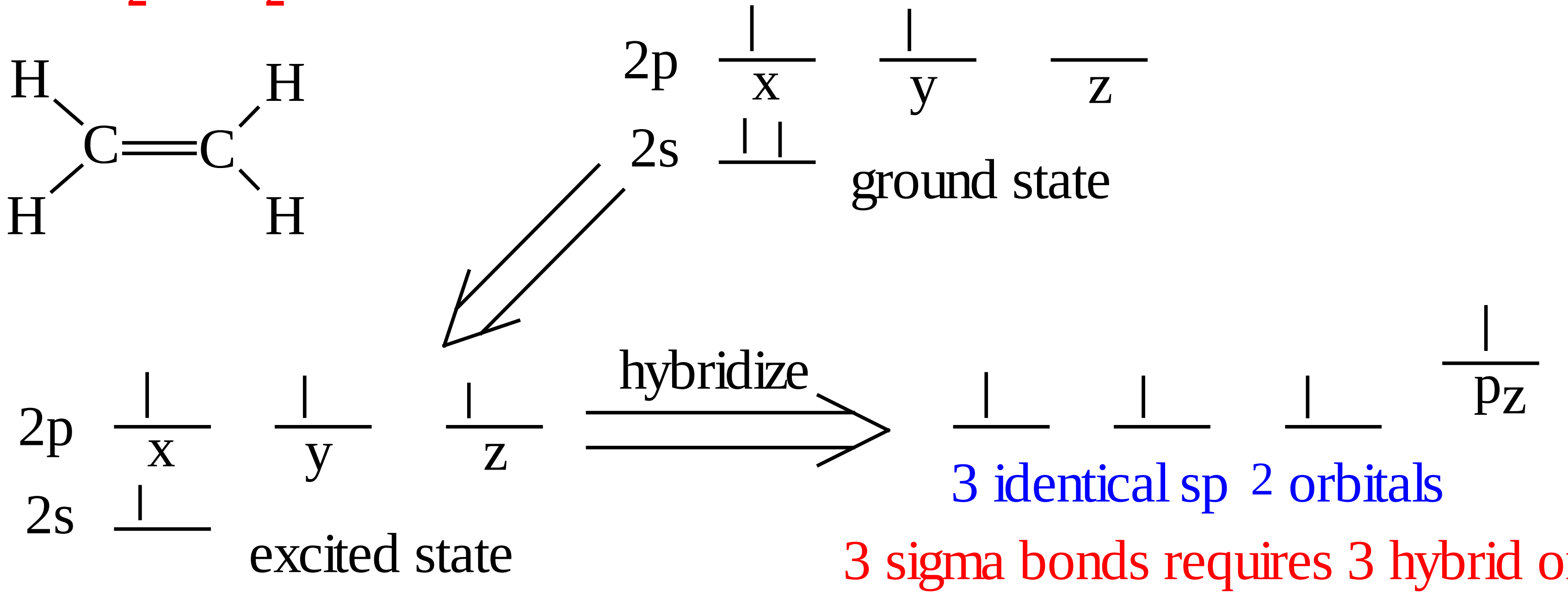
Ethylene $CH_2{=}CH_2$



sp^2 Hybridization

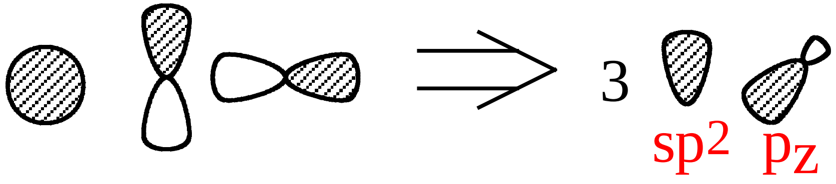
3 Regions of Electron Density

$CH_2{=}CH_2$

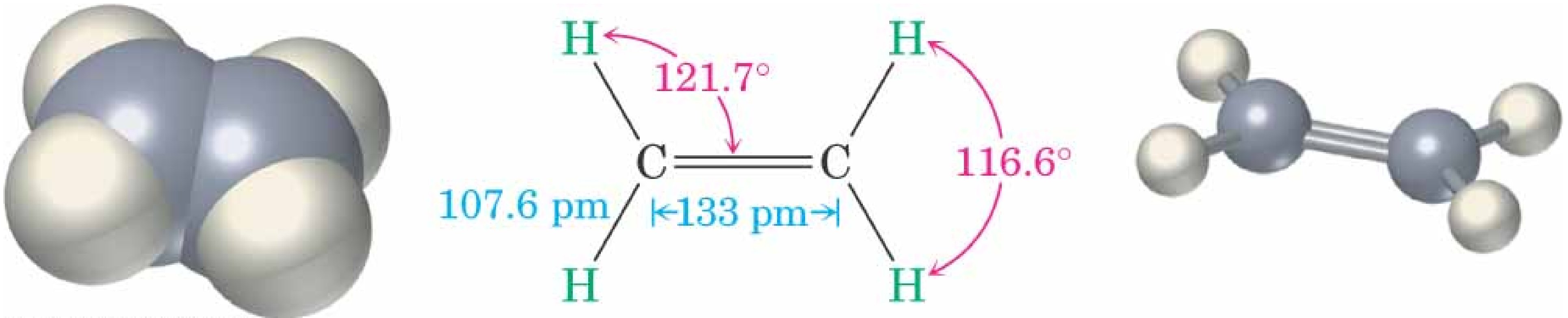


3 sigma bonds requires 3 hybrid orbitals

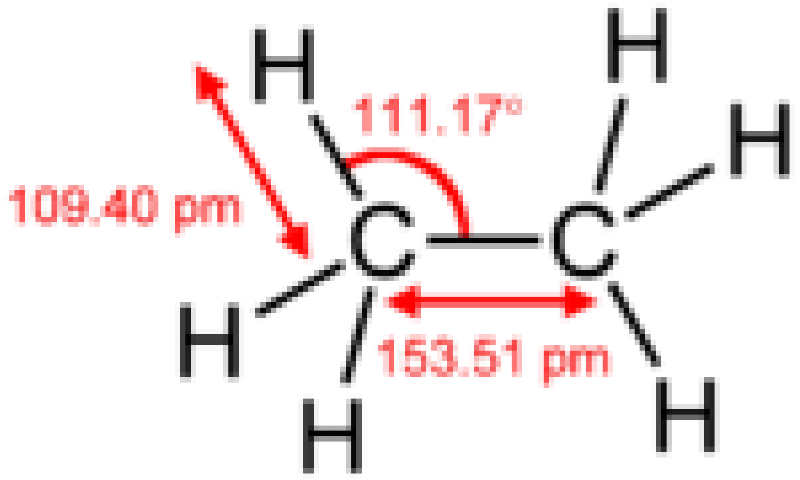
trigonal planar geometry

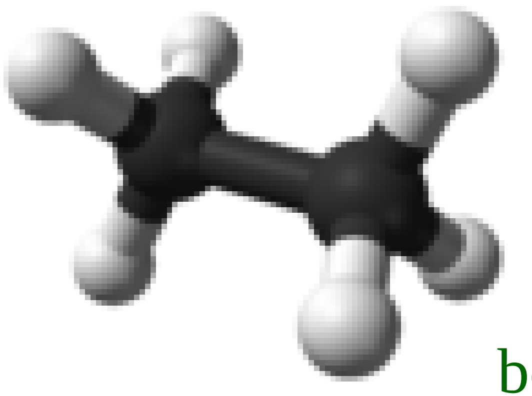


Ethylene



σ bond: ethane

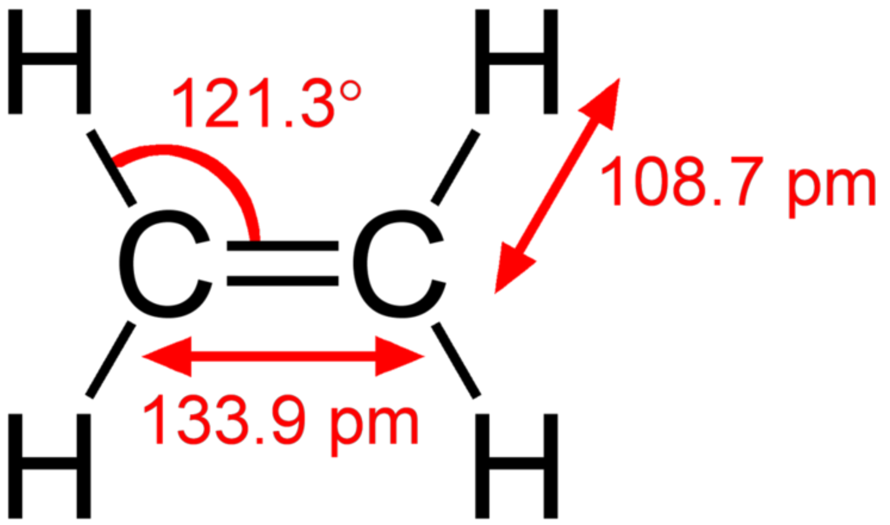


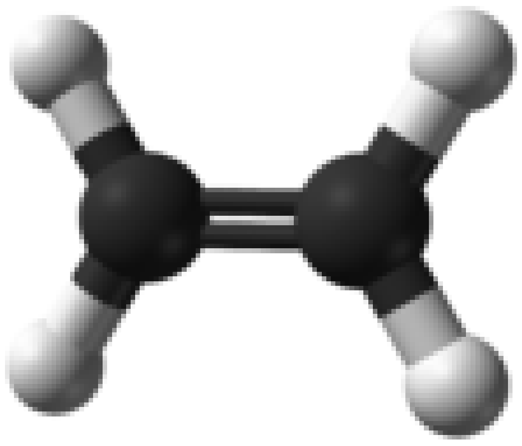


boiling point of ethene is **-88.6°C**

π bond: ethene

boiling point of ethene is **-103.7°C**





An Aliphatic Compound

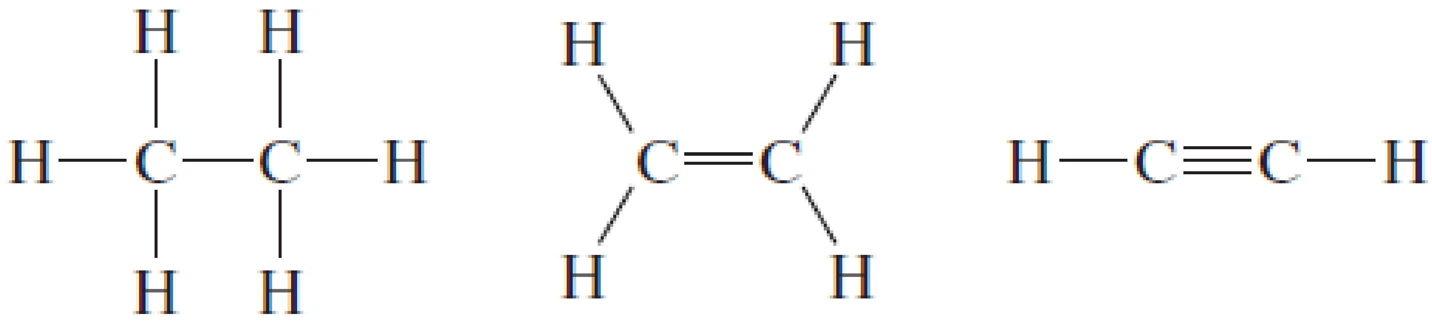
- Aliphatic hydrocarbons
- Aliphatic alcohol: alkanol
- Aliphatic carboxylic acid: alkanoic acid

CLASSES OF HYDROCARBONS

Hydrocarbons are compounds that contain only carbon and hydrogen and are divided into two main classes: **aliphatic** hydrocarbons and **aromatic** hydrocarbons

- Two sources were fats and oils, and the word *aliphatic* was derived from the Greek word *aleiphar* (“fat”).
- Aromatic hydrocarbons, irrespective of their own odor, were typically obtained by chemical treatment of pleasant-smelling plant extracts.

- Aliphatic hydrocarbons include three major groups: alkanes, alkenes, and alkynes.
- **Alkanes** are hydrocarbons in which all the bonds are single bonds, **alkenes** contain one or more carbon–carbon double bonds, and **alkynes** contain one or more carbon–carbon triple bonds.



Ethane (alkane) — Ethylene (alkene) — Acetylene (alkyne)

Ethane: a gas b.p. ~ -100°C

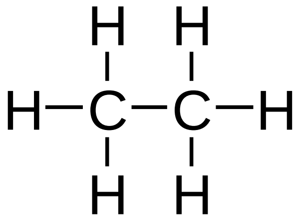
Empirical formula: CH_3

- An organic chemical
- Substance composed of organic molecules

Molecular formula C_2H_6

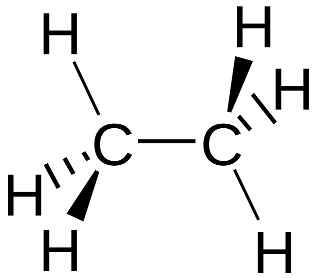
- Identity and number of atoms comprising each molecule

Structural formula



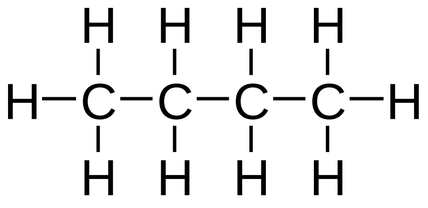
- Atom-to-atom connectivity

Structural formula showing stereochemistry



- 3D shape

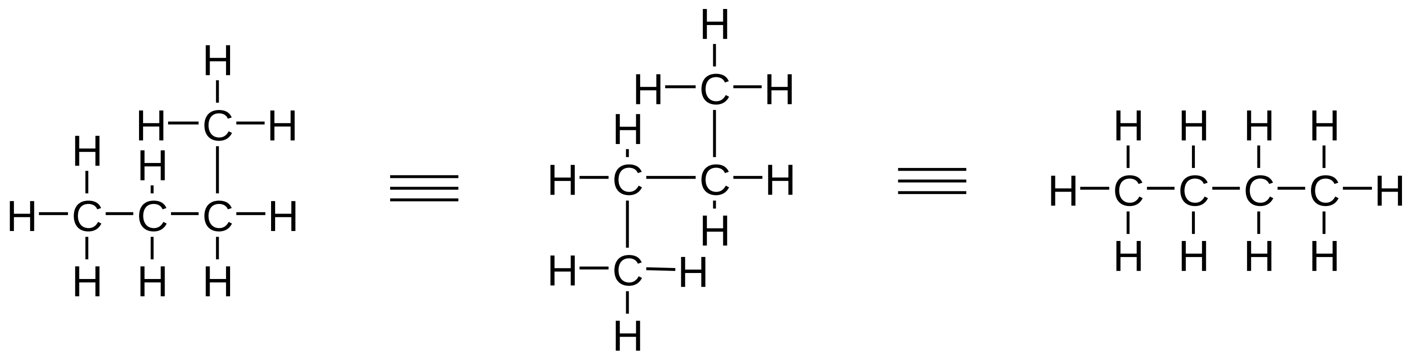
Representation and conformation



Butane
(full structural formula)

- Structural formulae: give information on *atom-to-atom connectivity*

- Do not give information on stereochemistry

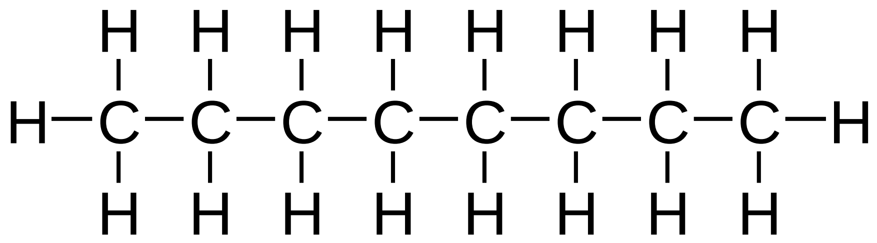


Same structural formula

Have the same information content

Representing larger molecules

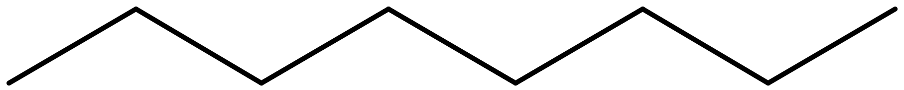
Full structural formula for, e.g. octane

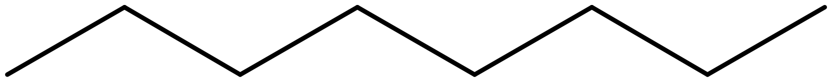


Condensed structural formula

$CH_3\text{-}CH_2\text{-}CH_2\text{-}CH_2\text{-}CH_2\text{-}CH_2\text{-}CH_2\text{-}CH_3$

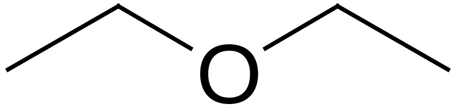
Line segment structural formula





Line segment structural formula for octane

- Each line represents a covalent bond between atoms
- Unless indicated otherwise, assume bonds are between Carbons
- C-H bonds not shown, assume they are present
- [so as make up valency of Carbon to 4]



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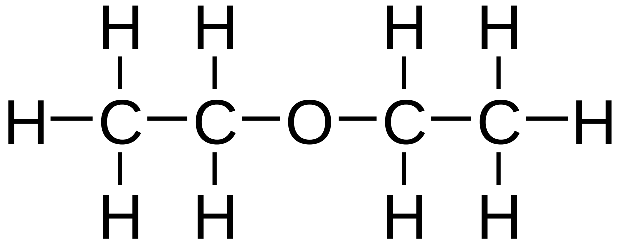
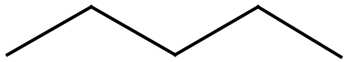
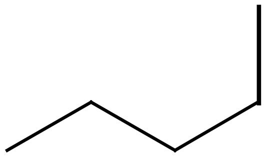
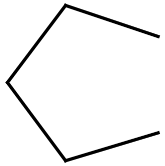
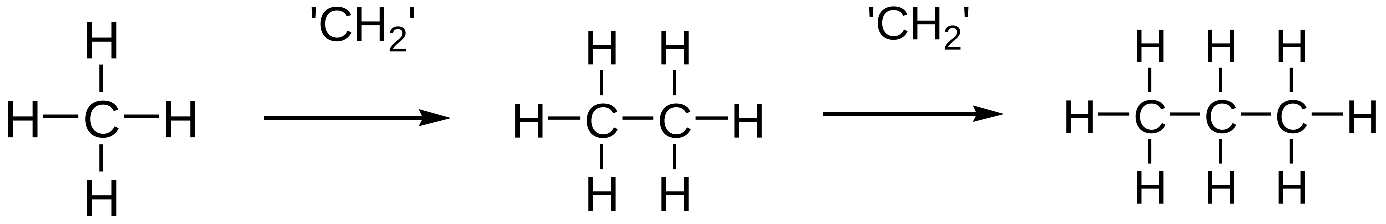
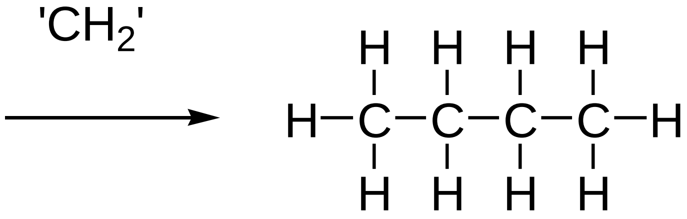


 =  =  etc. = pentane

Generating the series of alkanes by incrementally adding ‘CH_2’

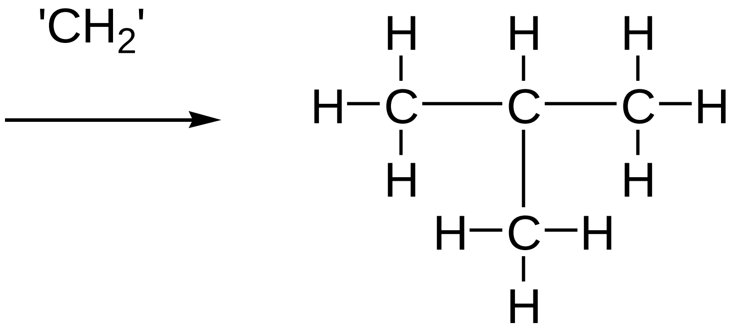


Methane Ethane Propane

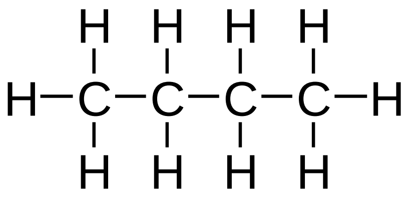


Butane

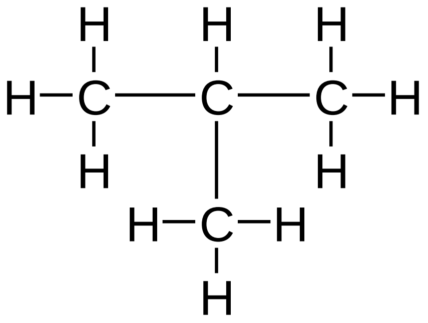
However, the last increment could also give



Isobutane



Butane (C_4H_{10})



Isobutane (C_4H_{10})

Structural isomers

'Isomer', from Greek *isos* (equal) and *meros* (in part)

•Structural isomers: same molecular formulae

•Different structural formulae
(different atom-to-atom connectivity)

- Structural isomers: different physical properties

CH_3-CH_2-CH_2-CH_3

n-butane

b.p. - 0.5^{o}C

$$\begin{array}{c} CH_3 \\ | \\ CH_3-CH-CH_3 \end{array}$$

isobutane

b.p. - 12.0^{o}C

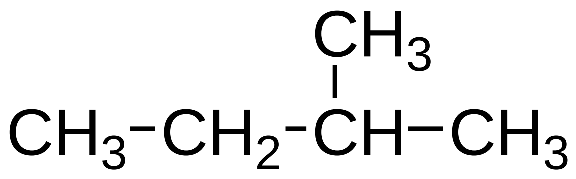
2-methylpropane

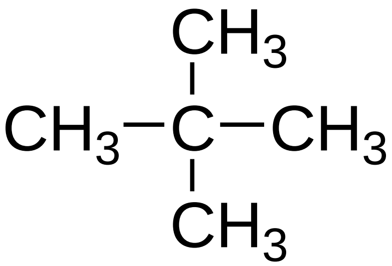
- Are different chemical entities

Isomers of C_4H_{10}. Both butane and 2-methylpropane have the same number of C and H atoms, but 2-methylpropane has a branched chain.

Pentane C_5H_{12} 3 structural isomers

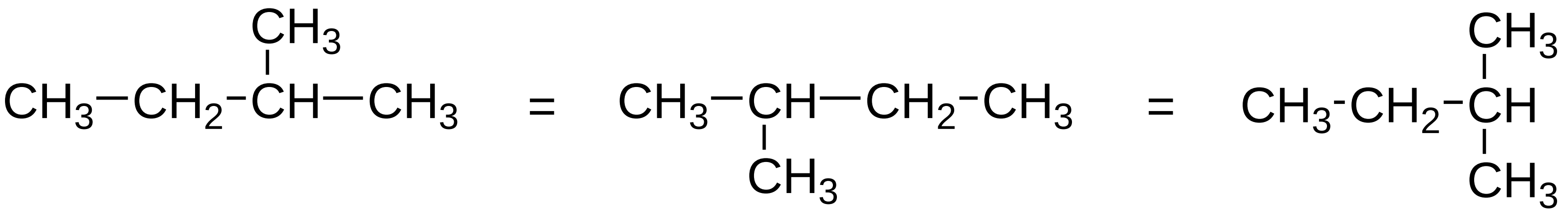
$CH_3-CH_2-CH_2-CH_2-CH_3$



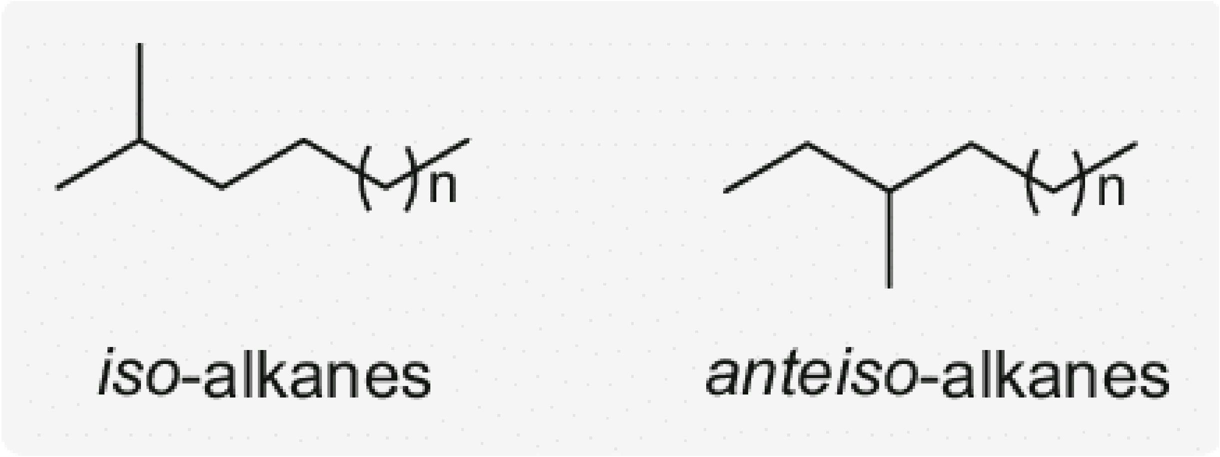


- All of these based on tetrahedral (**sp**$^{\mathbf{3}}$ hybridised) Carbon
- No other arrangements of C_5H_{12} possible

Note



etc.



Branched-chain fatty acids of the iso and anteiso series occur in many bacteria as the major acyl constituents of membrane lipids

Extent of structural isomerism in alkanes

Alkane	No. of structural isomers	
Methane	1	All known
Ethane	1	
Propane	1	
Butane	2	
Pentane	3	
Hexane	5	
Decane	75	
Pentadecane	4347	
Eicosane	366,319	
Triacontane ($C_{30}H_{62}$)	44×10^9	

How to name them?

IUPAC name

In chemical nomenclature, the **IUPAC nomenclature of organic chemistry** is a systematic method of naming organic chemical compounds as recommended by the International Union of Pure and Applied Chemistry (IUPAC).

It is published in the *Nomenclature of Organic Chemistry* (informally called the Blue Book). Ideally, every possible organic compound should have a name from which an unambiguous structural formula can be created. There is also an IUPAC nomenclature of inorganic chemistry.

Need to expand the system of **nomenclature** to allow naming of individual structural isomers

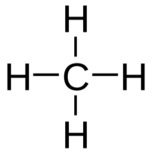
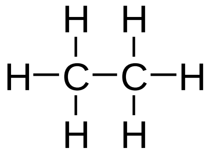
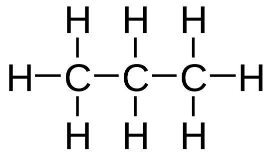
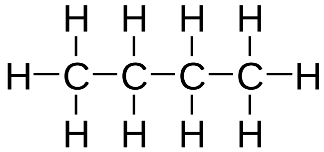
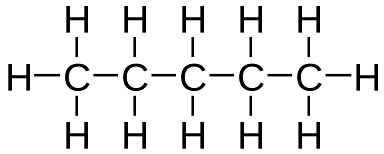
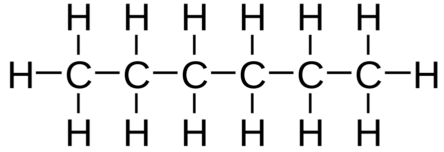
- Compounds without branches are called ‘straight chain’

- Branched compounds are named as ***alkyl*** derivatives of the longest straight chain in the molecule

- The length of the longest chain provides the parent name

- The straight chain is numbered to allow indication of the point of branching

- The branching *alky*l groups (or ***substituents***) are named from the corresponding alkane

n	Molecular Formula	Structural formula	Name	Condensed structural formula
1	CH_4		methane	CH_4
2	C_2H_6		ethane	CH_3CH_3
3	C_3H_8		propane	$CH_3CH_2CH_3$
4	C_4H_{10}		butane	$CH_3CH_2CH_2CH_3$
5	C_5H_{12}		pentane	$CH_3CH_2CH_2CH_2CH_3$
6	C_6H_{14}		hexane	$CH_3CH_2CH_2CH_2CH_2CH_3$

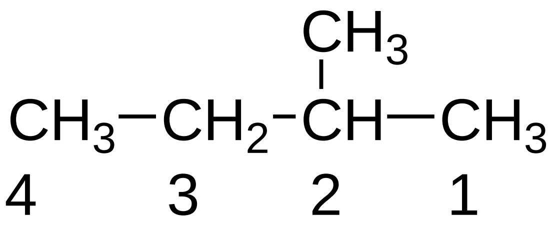
Further members of the series

Heptane	$CH_3CH_2CH_2CH_2CH_2CH_2CH_3$
Octane	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_3$
Nonane	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_3$
Decane	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_3$
Undecane	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_3$
Dodecane	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_3$

Etc., etc.

Alkane	Alkyl group
Methane	Methyl (CH_3-)
Ethane	Ethyl (CH_3CH_2-)
Propane	Propyl ($CH_3CH_2CH_2$-)
Butane	Butyl ($CH_3CH_2CH_2CH_2$-)

Etc.



2-Methylbutane

[Straight chain numbered so as to give the lower branch number]

Linear Alkyl Chains

# of Carbons	Base Name	Substituent	Prefix
1	methane	methyl	
2	ethane	ethyl	di-
3	propane	propyl (isopropyl)	tri-
4	butane	butyl (isobutyl, t-butyl)	tetra-
5	pentane	pentyl	penta-
6	hexane	hexyl	hexa-
7	heptane	heptyl	hepta-
8	octane	octyl	octa-
9	nonane	nonyl	nona-
10	decane	decyl	deca-
11	undecane	undecyl-	
12	dodecane	dodecyl-	
13	tridecane	tridecyl-	
20	icosane	icosyl-	
21	henicosane	henicosyl-	
22	docosane	docosyl-	
23	tricosane	tricosyl-	
30	triacontane	triacontyl-	
31	untriacontane	untriacontyl-	
32	dotriacontane	dotriacontyl-	
33	tritriacontane	tritriacontyl-	
40	tetradecane	tetradecyl-	
50	pentadecane	pentadecyl-	

The prefixes are used to describe multiple substituents of the same type (ie, trimethylhexane, octahydronaphthalene).

Some points concerning this series of alkanes

1. Series is generated by repeatedly adding ‘CH_2’ to the previous member of the series

A series generated in this manner is known as an ***homologous series***

2. **Nomenclature** (naming)

Names all share a common suffix, i.e.’ …ane’

The suffix ‘…ane’ indicates that the compound is an alkane

The prefix indicates the *number of carbons in the compound*

‘Meth…’ = 1 Carbon

‘Eth…’ = 2 Carbons

‘Prop…’ = 3 Carbons

‘But…’ = 4 Carbons

‘Pent…’ = 5 Carbons

‘Hex…’ = 6 Carbons

‘Hept…’ = 7 Carbons

‘Oct…’ = 8 Carbons

‘Non…’ = 9 Carbons

‘Dec…’ = 10 Carbons

‘Undec…’ = 11 Carbons

‘Dodec…’ = 12 Carbons

Heptane $\mathbf{CH_3CH_2CH_2CH_2CH_2CH_2CH_3}$

‘Hept…’ implies 7 Carbons

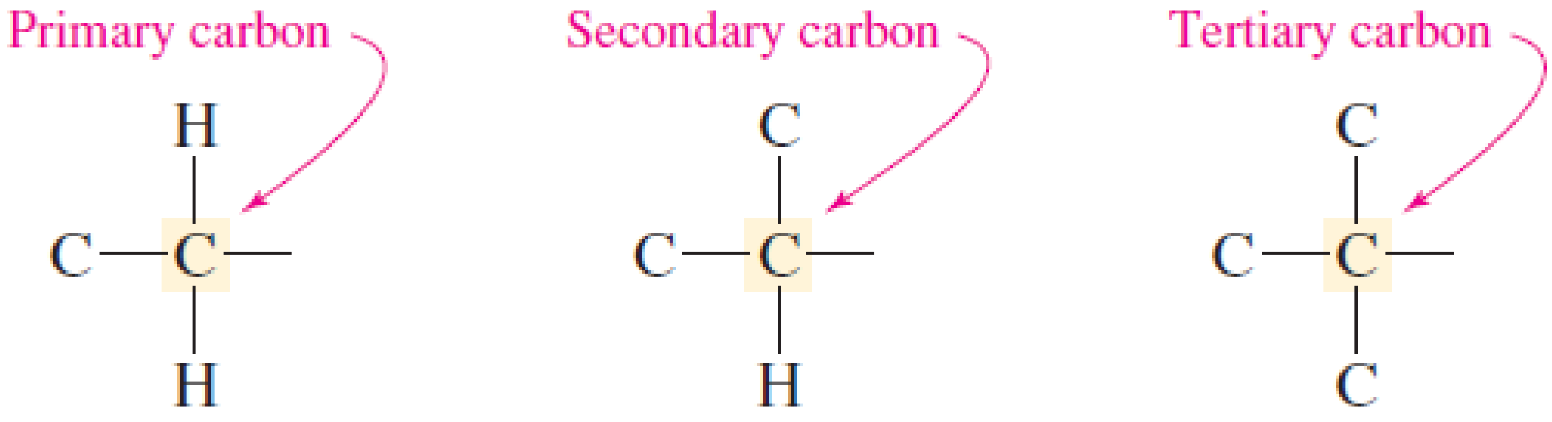
‘…ane’ implies compound is an alkane

Number of Carbon Atoms	Name	Number of Carbon Atoms	Name
1	Methane	11	Undecane
2	Ethane	12	Dodecane
3	Propane	13	Tridecane
4	Butane	14	Tetradecane
5	Pentane	15	Pentadecane
6	Hexane	16	Hexadecane
7	Heptane	17	Heptadecane
8	Octane	18	Octadecane
9	Nonane	19	Nonadecane
10	Decane	20	Icosane*

* Spelled “eicosane” prior to 1979 version of IUPAC rules.

Carbon atoms are classified according to their degree of substitution by other carbons.

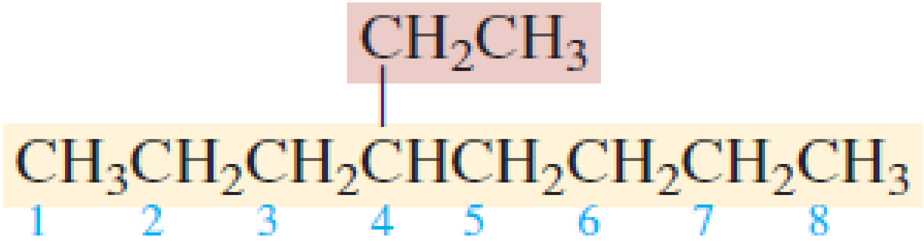
- A **primary** carbon is *directly* attached to one other carbon.
- Similarly, a **secondary** carbon is directly attached to two other carbons, a **tertiary** carbon to three, and a **quaternary** carbon to four.
- Alkyl groups are designated as primary, secondary, or tertiary according to the degree of substitution of the carbon at the potential point of attachment.



Primary alkyl group　　Secondary alkyl group　　Tertiary alkyl group

IUPAC NAMES OF HIGHLY BRANCHED ALKANES

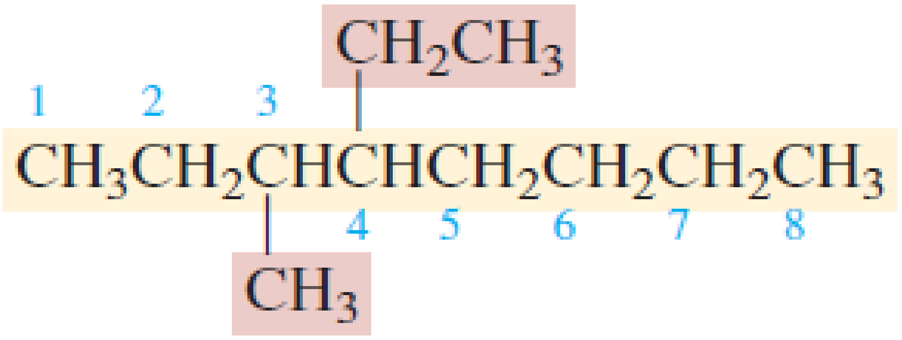
- By combining the basic principles of IUPAC notation with the names of the various alkyl groups, we can develop systematic names for highly branched alkanes.



4-ethyloctane

What happens to the IUPAC name when a methyl replaces one of the hydrogens at C-3?

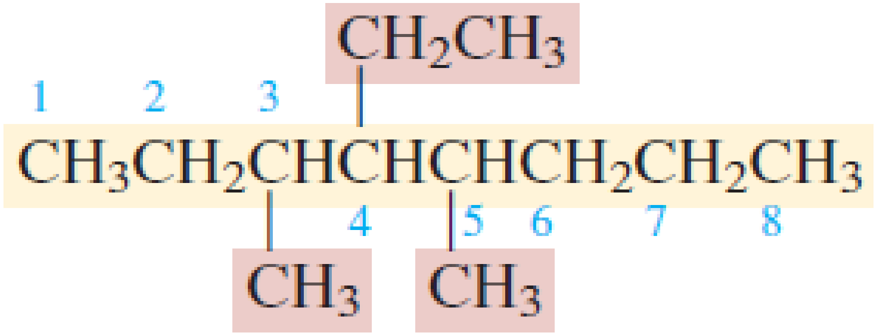
The compound becomes an octane derivative that bears a C-3 methyl group and a C-4 ethyl group.

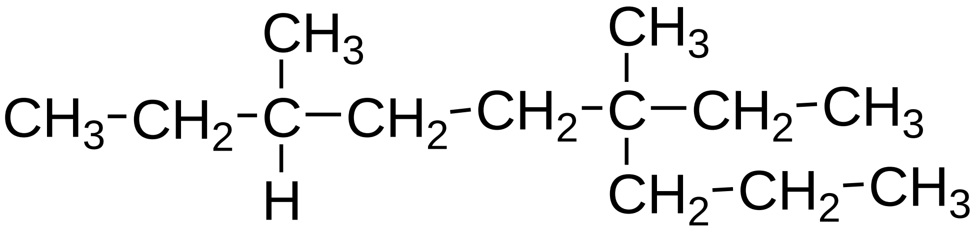


When two or more different substituents are present, they are listed in alphabetical order in the name.

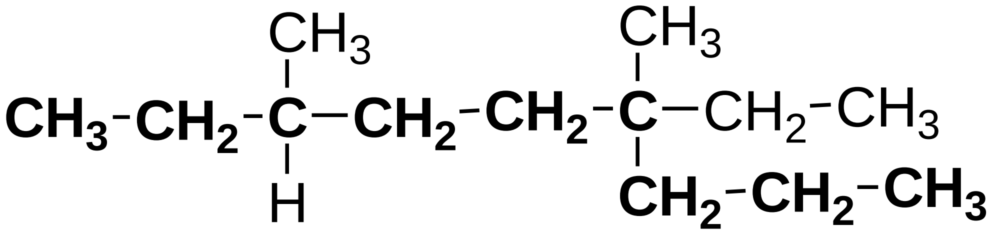
The IUPAC name for this compound is **4-ethyl-3-methyloctane.**

- Replicating prefixes such as *di*-, *tri*-, and *tetra*- are used as needed but are ignored when alphabetizing.
- Adding a second methyl group to the original structure, at C-5, for example, converts it to **4-ethyl-3,5-dimethyloctane.**



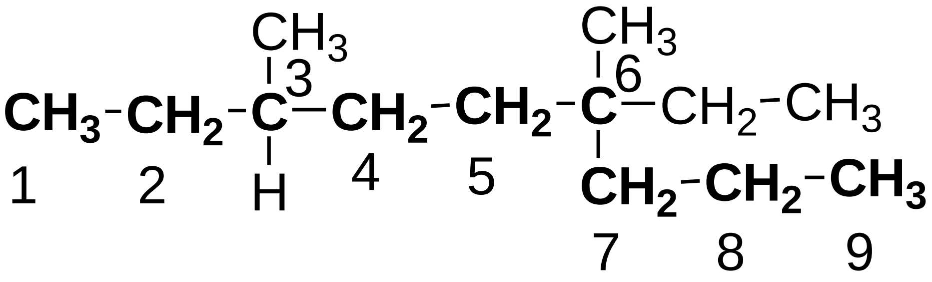


First, identify longest straight chain



‘…nonane’

Number so as to give lower numbers for branch points



Branches at C3 and C6
Not at C4 and C7

6-ethyl-3,6-dimethyl-nonane

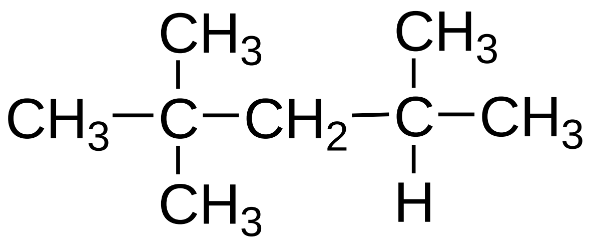
Identical substituents grouped together with a prefix

•'di…' for two identical

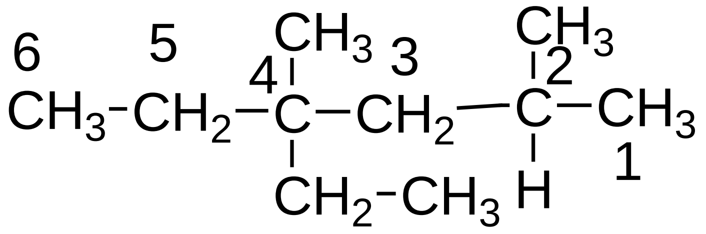
•'tri…' for three

•'tetra…' for four

Substituents named in alphabetical order



2,2,4-Trimethylpentane



4-ethyl-2,4-dimethylhexane

LPG: Propane (propylene, butane, and butylene))

Gasoline: Octane

Kerosene: $C_{12}H_{26}$

Diesel: $C_{16}H_{34}$

Heptane, iso-octane

RON, MON, AKI

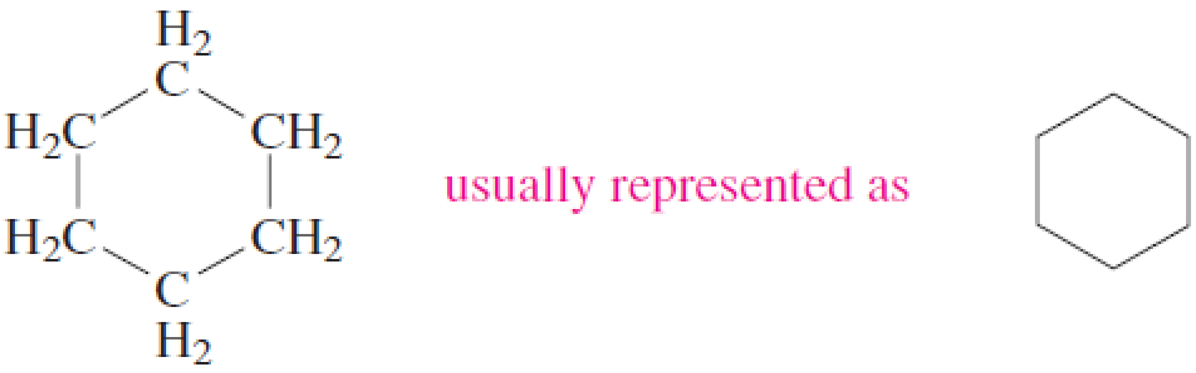
BIG DIFFERENCE BETWEEN straight *n*-alkane

and branched alkane

CYCLOALKANE NOMENCLATURE

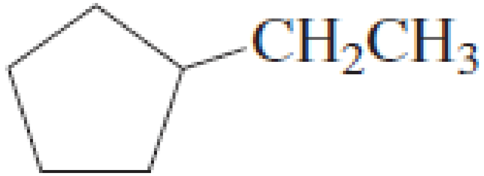
Cycloalkanes are alkanes that contain a ring of three or more carbons. They are frequently encountered in organic chemistry and are characterized by the molecular formula C*n*H2*n*.

Cyclopropane

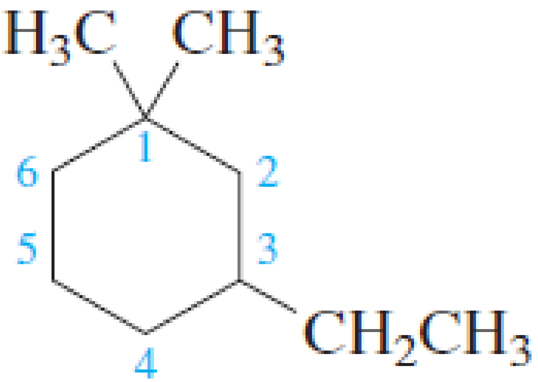


Cyclohexane

- cycloalkanes are named, under the IUPAC system, by adding the prefix *cyclo*- to the name of the unbranched alkane with the same number of carbons as the ring.
- Attached groups are identified in the usual way. Their positions are specified by numbering the carbon atoms of the ring in the direction that gives the lowest number to the substituted carbon at the first point of difference.

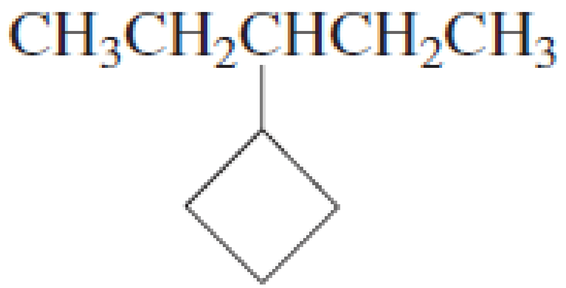


Ethylcyclopentane



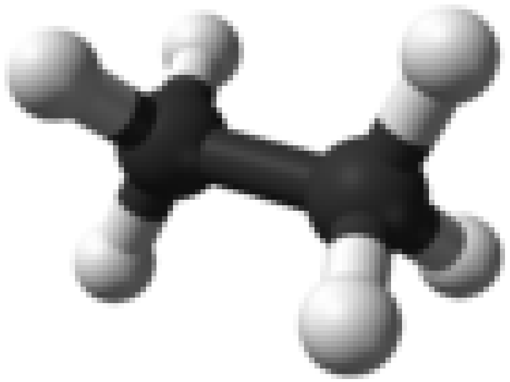
3-Ethyl-1,1-dimethylcyclohexane
(not 1-ethyl-3,3-dimethylcyclohexane, because first point of difference rule requires 1,1,3 substitution pattern rather than 1,3,3)

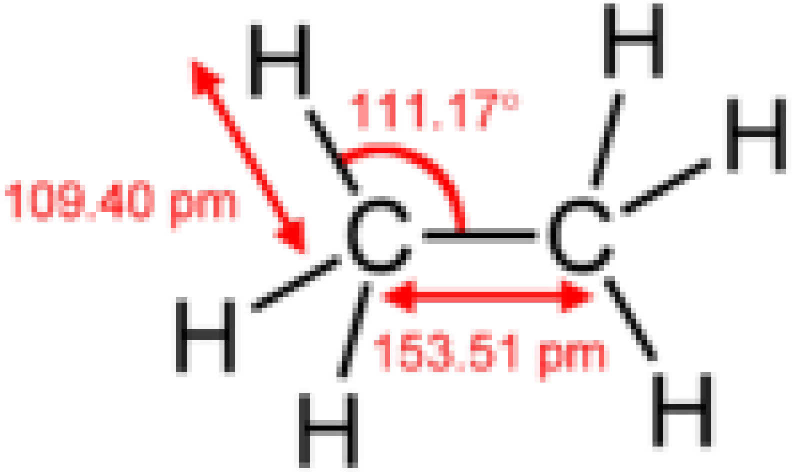
When the ring contains fewer carbon atoms than an alkyl group attached to it, the compound is named as an alkane, and the ring is treated as a cycloalkyl substituent:



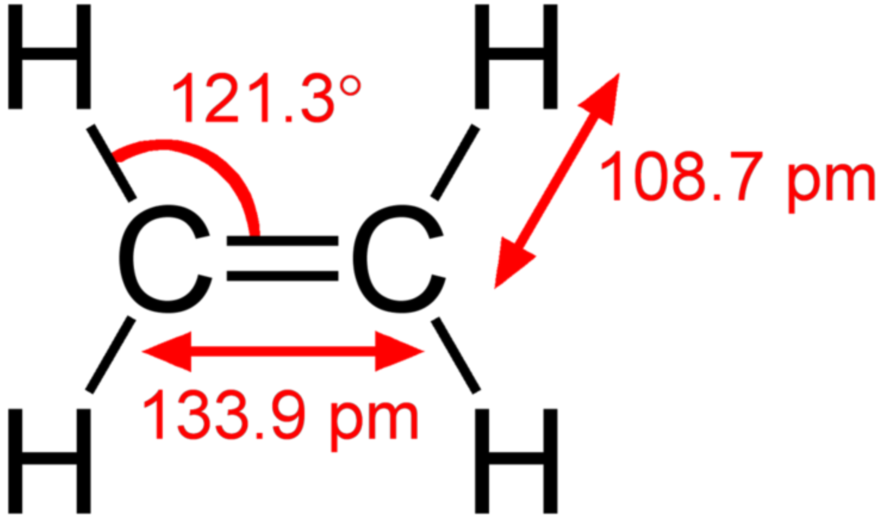
3-Cyclobutylpentane

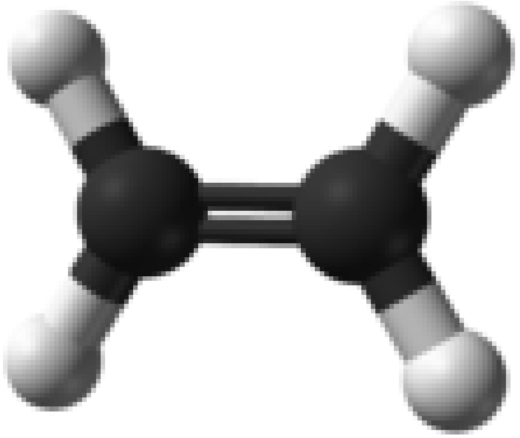
σ bond: ethane



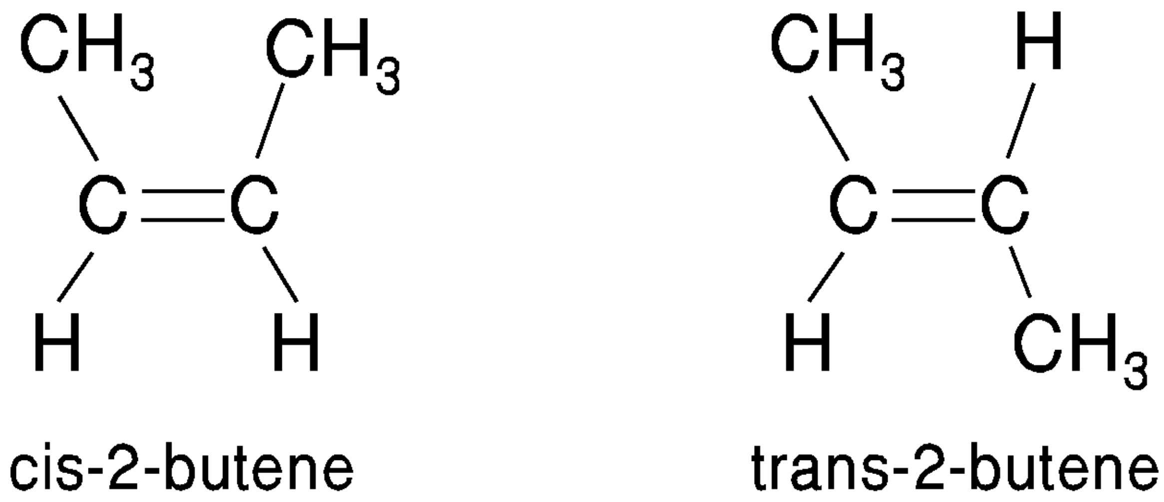


π bond: ethene



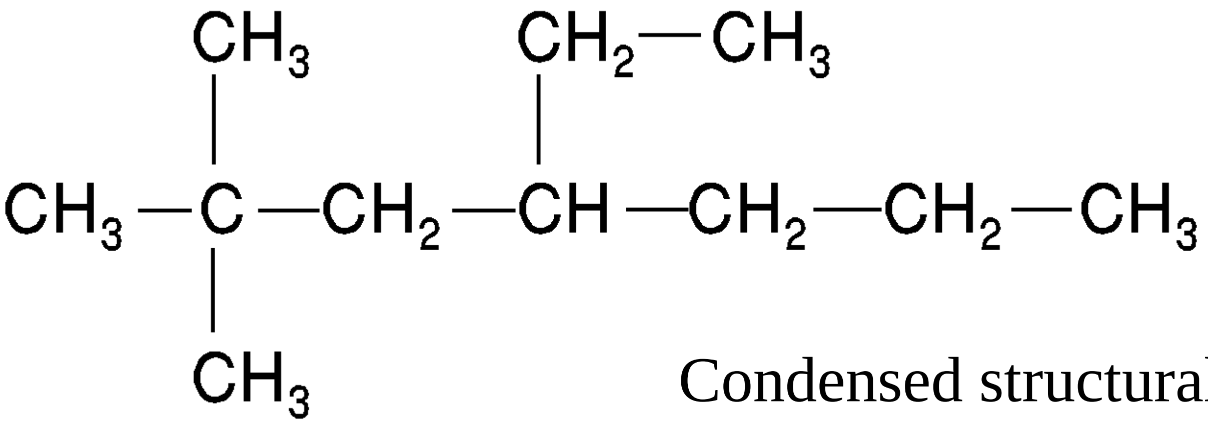


Cis and trans configuration



Examples of geometric isomers, *cis*-2-butene and *trans*-2-butene. In the first isomer, the two substituent groups are on the same side of the double bond. In the second isomer, the two substituent groups are on opposite sides of the double bond.

Condensed structural formula

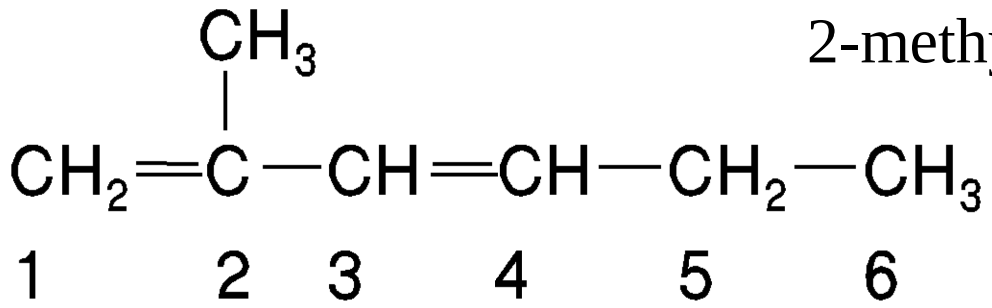


Condensed structural formula for the alkane 4-ethyl-2,2-dimethylheptane.

$$\underset{6}{CH_3}-\underset{5}{CH_2}-\underset{4}{CH_2}-\underset{3}{CH}=\underset{2}{CH}-\underset{1}{CH_3}$$

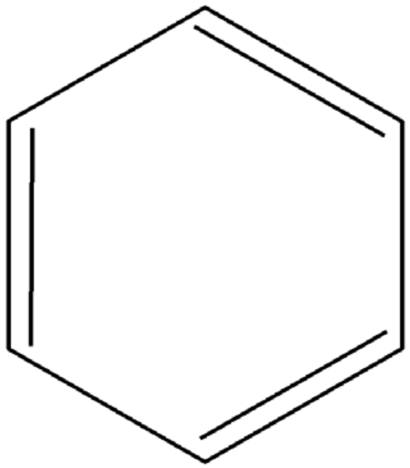
Condensed structural formula for 2-hexene.

Condensed structural formula

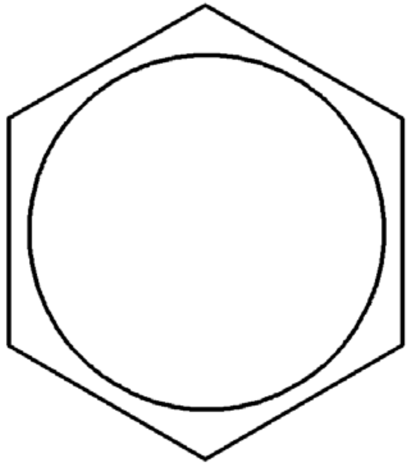


2-methyl-1,3-hexadiene

Benzene



Benzene ring shown as alternating single and double bonds



Benzene ring shown as a ring of delocalized electrons

Benzene is much more stable than would be expected if we treated each double bond independently. This is because all six pi orbitals overlap, and the resulting hybrid molecular orbitals are more stable than an independent pi orbital. This is one definition of an “aromatic” molecule.

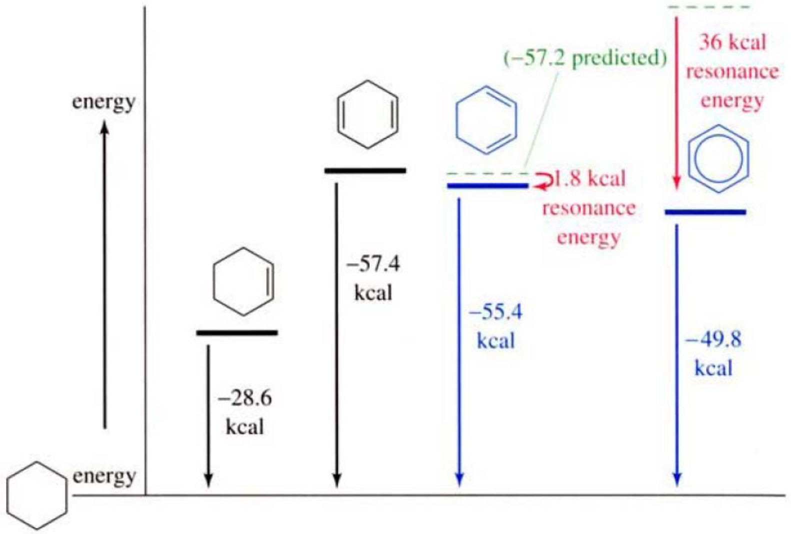
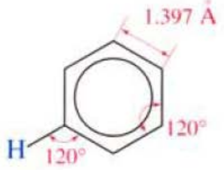
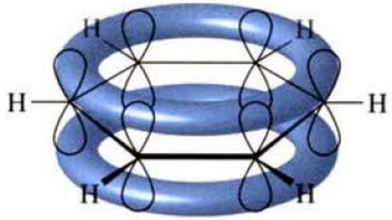
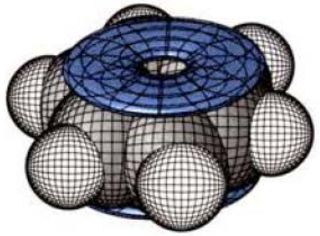


FIGURE 16-2 The molar heats of hydrogenation and the relative energies of cyclohexene, 1,4-cyclohexadiene, 1,3-cyclohexadiene, and benzene. The dashed lines represent the energies that would be predicted if every double bond had the same energy as the double bond in cyclohexene.

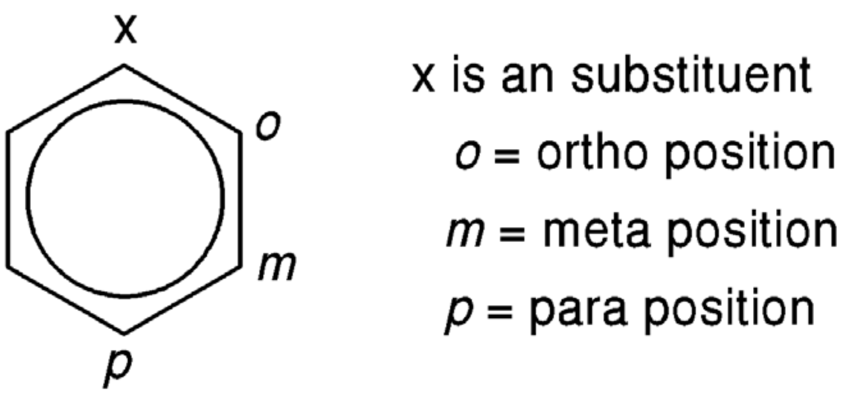






Wade, Organic Chemistry

Ring attachments



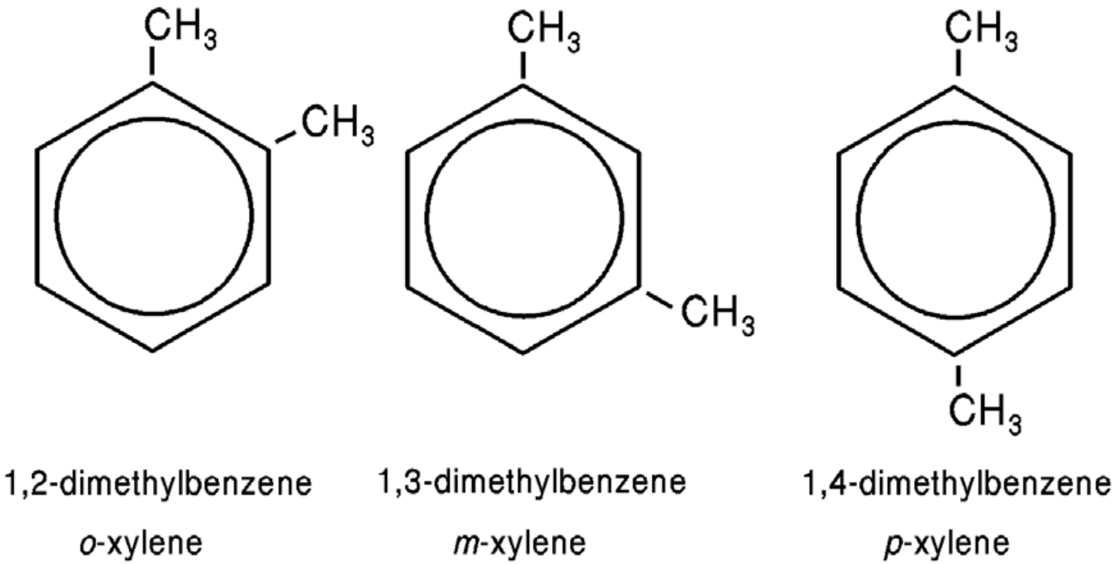
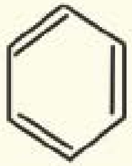
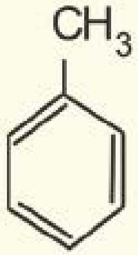


Illustration of the naming conventions for the benzene ring using attached methyl groups. Note that while not shown on the diagram, the other sites are occupied by hydrogen atoms.

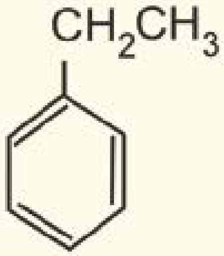
Organic Pollutants: BTEX Compounds



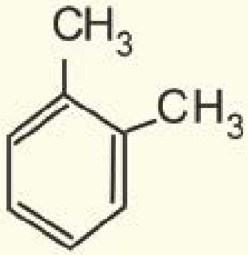
benzene



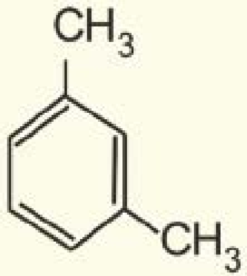
toluene



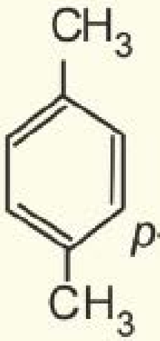
ethyl-
benzene



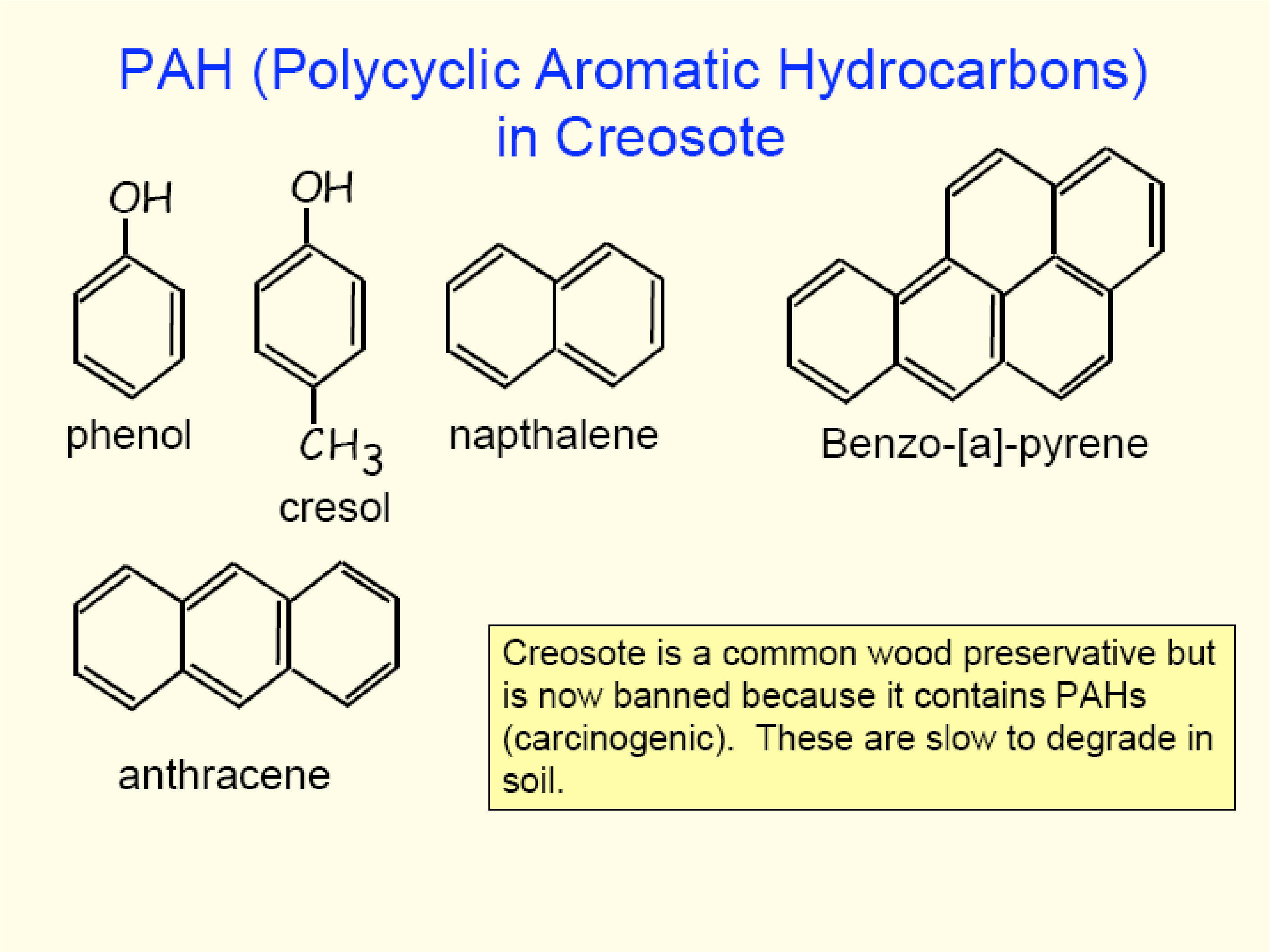
o-xylene

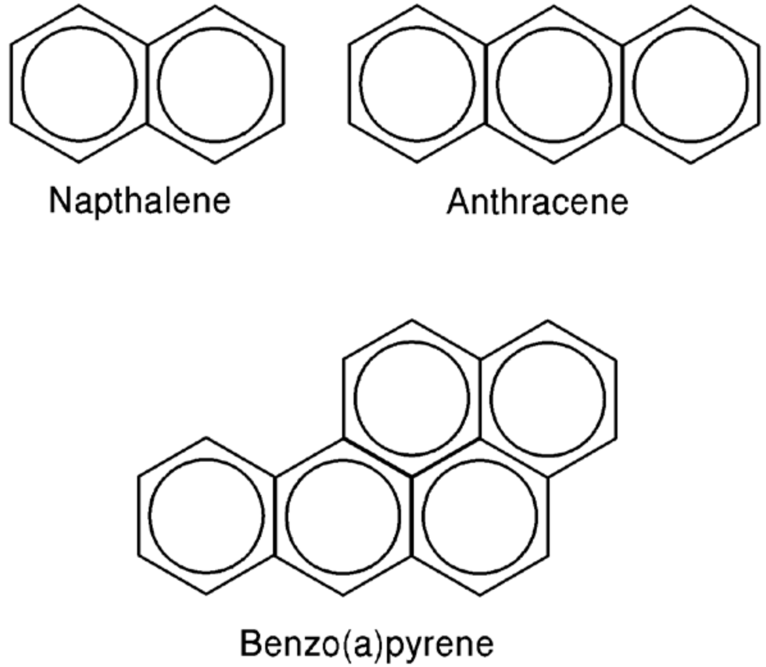


m-xylene



p-xylene





Examples of polycyclic aromatic hydrocarbons (PAHs) including the know carcinogen benzo(a)pyrene. Where the benzene rings do not share edges, hydrogen atoms are attached to the carbon atoms.

Compounds containing O, N, and S

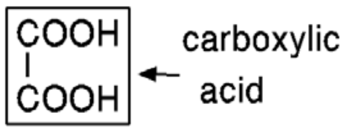
Functional Groups in Some Important Classes of Organic Compounds

Class	Generalized Abbreviation	Representative Example	Name of Example*
Alcohol	ROH	CH_3CH_2OH	Ethanol
Alkyl halide	RCl	CH_3CH_2Cl	Chloroethane
Amine†	RNH_2	$CH_3CH_2NH_2$	Ethanamine
Epoxide	$R_2C\underset{O}{\text{—}}CR_2$	$H_2C\underset{O}{\text{—}}CH_2$	Oxirane
Ether	ROR	$CH_3CH_2OCH_2CH_3$	Diethyl ether
Nitrile	$RC{\equiv}N$	$CH_3CH_2C{\equiv}N$	Propanenitrile
Nitroalkane	RNO_2	$CH_3CH_2NO_2$	Nitroethane
Thiol	RSH	CH_3CH_2SH	Ethanethiol

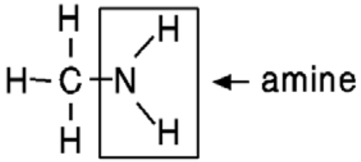
Classes of Compounds That Contain a Carbonyl Group

Class	Generalized Abbreviation	Representative Example	Name of Example
Aldehyde	$R\overset{O}{\overset{\|}{C}}H$	$CH_3\overset{O}{\overset{\|}{C}}H$	Ethanal
Ketone	$R\overset{O}{\overset{\|}{C}}R$	$CH_3\overset{O}{\overset{\|}{C}}CH_3$	2-Propanone
Carboxylic acid	$R\overset{O}{\overset{\|}{C}}OH$	$CH_3\overset{O}{\overset{\|}{C}}OH$	Ethanoic acid
Carboxylic acid derivatives:			
Acyl halide	$R\overset{O}{\overset{\|}{C}}X$	$CH_3\overset{O}{\overset{\|}{C}}Cl$	Ethanoyl chloride
Acid anhydride	$R\overset{O}{\overset{\|}{C}}O\overset{O}{\overset{\|}{C}}R$	$CH_3\overset{O}{\overset{\|}{C}}O\overset{O}{\overset{\|}{C}}CH_3$	Ethanoic anhydride
Ester	$R\overset{O}{\overset{\|}{C}}OR$	$CH_3\overset{O}{\overset{\|}{C}}OCH_2CH_3$	Ethyl ethanoate
Amide	$R\overset{O}{\overset{\|}{C}}NR_2$	$CH_3\overset{O}{\overset{\|}{C}}NH_2$	Ethanamide

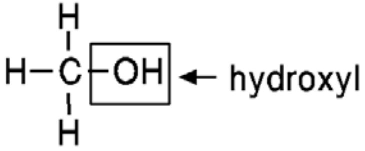
Functional groups



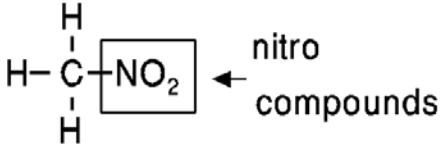
Oxalic acid



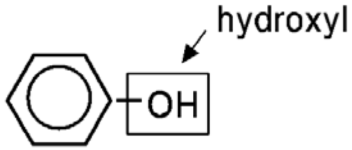
Methylamine



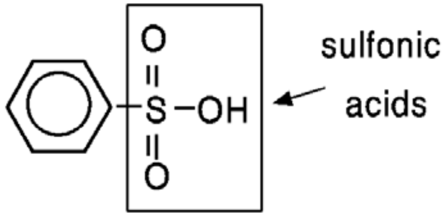
Methanol



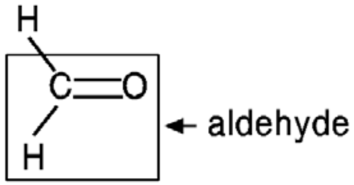
Nitromethane



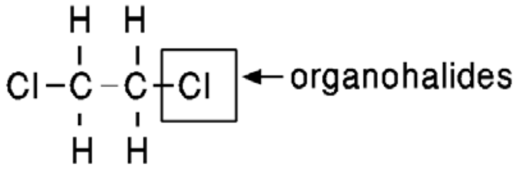
Phenol



Benzenesulfonic acid



Formaldehyde



1,2-Dichloroethane

Rules for naming the alcohols

1. Find the longest chain containing the hydroxy group (OH). If there is a chain with more carbons than the one containing the OH group it will be named as a subsitutent.

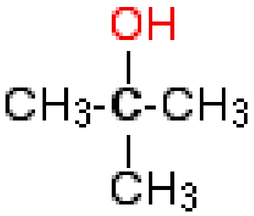
2.Place the OH on the lowest possible number for the chain. With the exception of carbonyl groups such as ketones and aldehydes, the alcohol or hydroxy groups have first priority for naming.

3.When naming a cyclic structure, the -OH is assumed to be on the first carbon unless the carbonyl group is present, in which case the later will get priority at the first carbon.

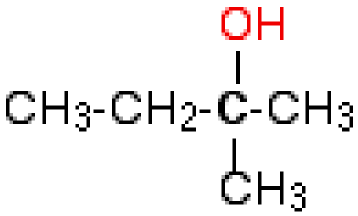
4.When multiple -OH groups are on the cyclic structure, number the carbons on which the -OH groups reside

5.Remove the final **e** from the parent **alkane** chain and add **-ol**. When multiple alcohols are present use **di, tri,** et.c before the **ol,** after the parent name. ex. 2,3-hexan**diol.** If a carbonyl group is present, the -OH group is named with the prefix "hydroxy," with the carbonyl group attached to the parent chain name so that it ends with **-al** or **-one.**

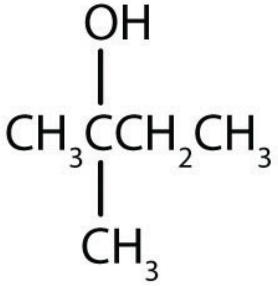
Name	Molecular Formula
Methanol (methyl alcohol)	CH_3OH
Ethanol (ethyl alcohol)	C_2H_5OH
Propanol	C_3H_7OH
Butanol	C_4H_9OH
Pentanol	$C_5H_{11}OH$
Hexanol	$C_6H_{13}OH$
Heptanol	$C_7H_{15}OH$
Octanol	$C_8H_{17}OH$



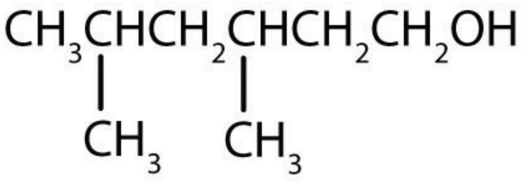
2-methylpropan-2-ol



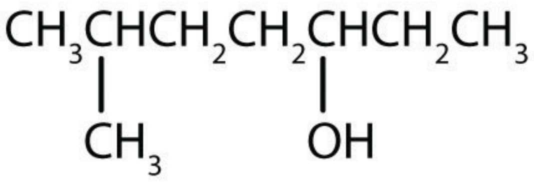
2-methylbutan-2-ol



2-methyl-2-butanol



3,5-dimethyl-1-hexanol



6-methyl-3-heptanol

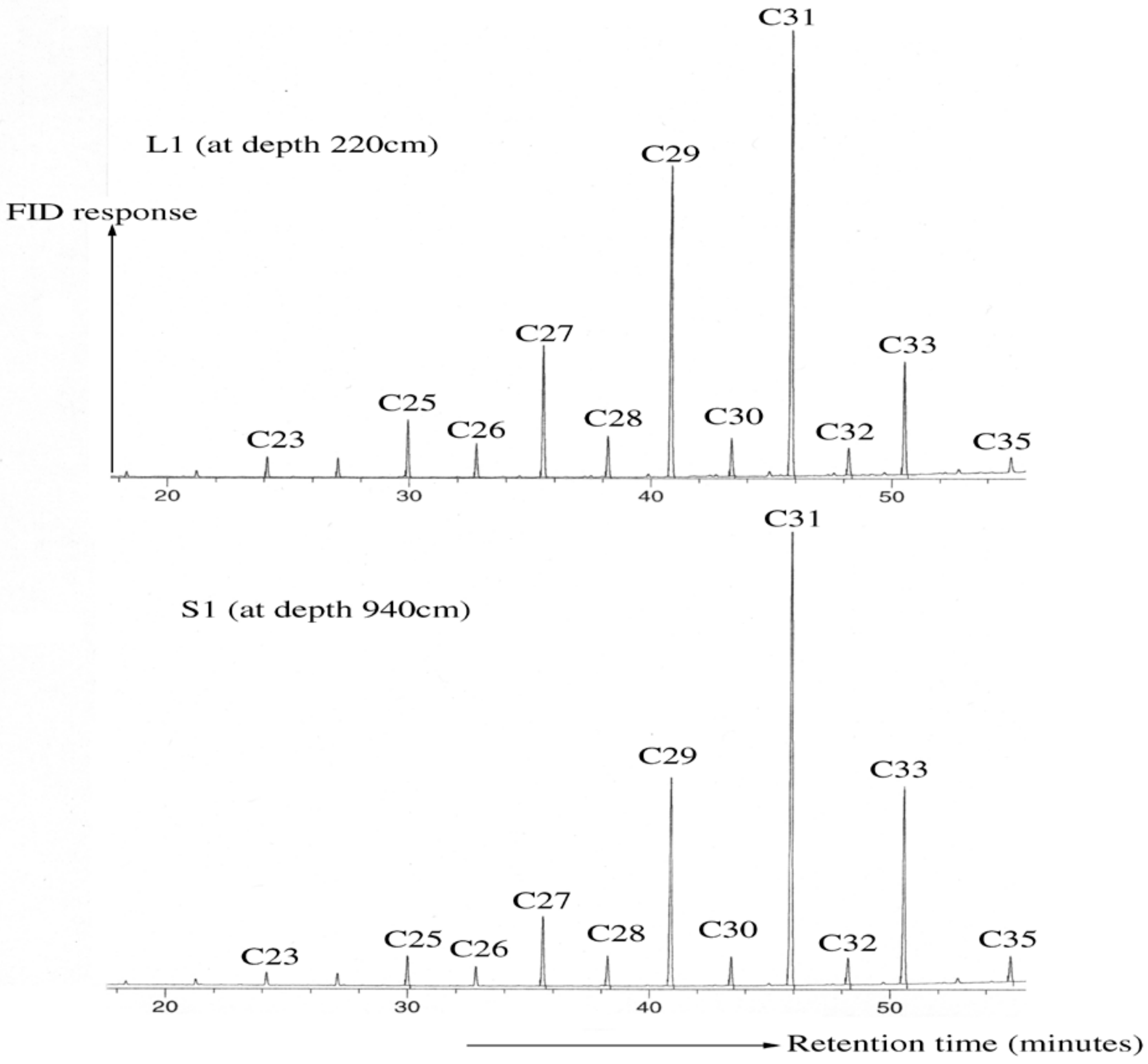
Rules 1 and 2

Fatty alcohols **(or long-chain alcohols) are usually high-molecular-weight, straight-chain primary alcohols**

- 1-Triacontanol
- **Triacontan-1-ol**
- Triacontanol

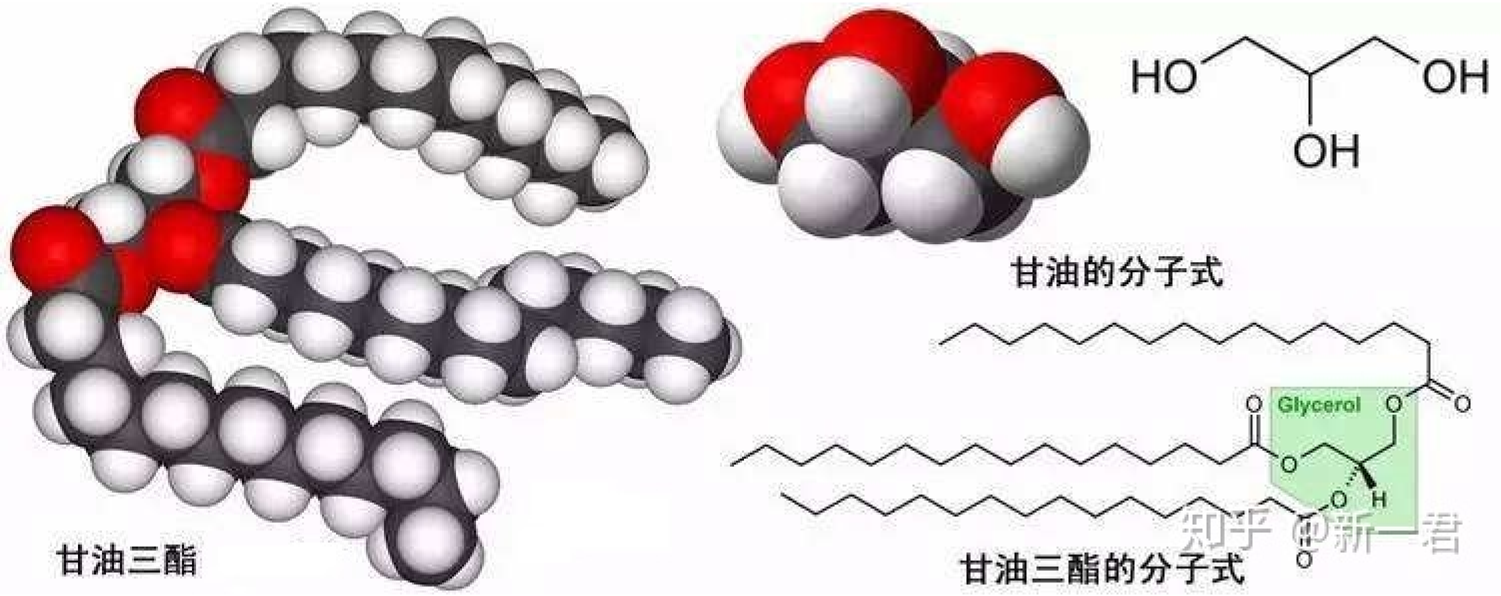
438.8 g/mol

Nonacosane



GC trace of *n*-alkane fraction of extraction from loess (L1, top)
and paleosol (S2, bottom)

Fatty acids (FA) are carboxylic acids with long aliphatic chains which may be straight or branched, saturated or unsaturated



Nomenclature of Carboxylic Acids

- The IUPAC system of nomenclature assigns a characteristic suffix to these classes. The **–e** ending is removed from the name of the parent chain and is replaced **-anoic acid**.

- Since a carboxylic acid group must always lie at the end of a carbon chain, it is always is given the #1 location position in numbering and it is not necessary to include it in the name.

n-Octadecanoic acid

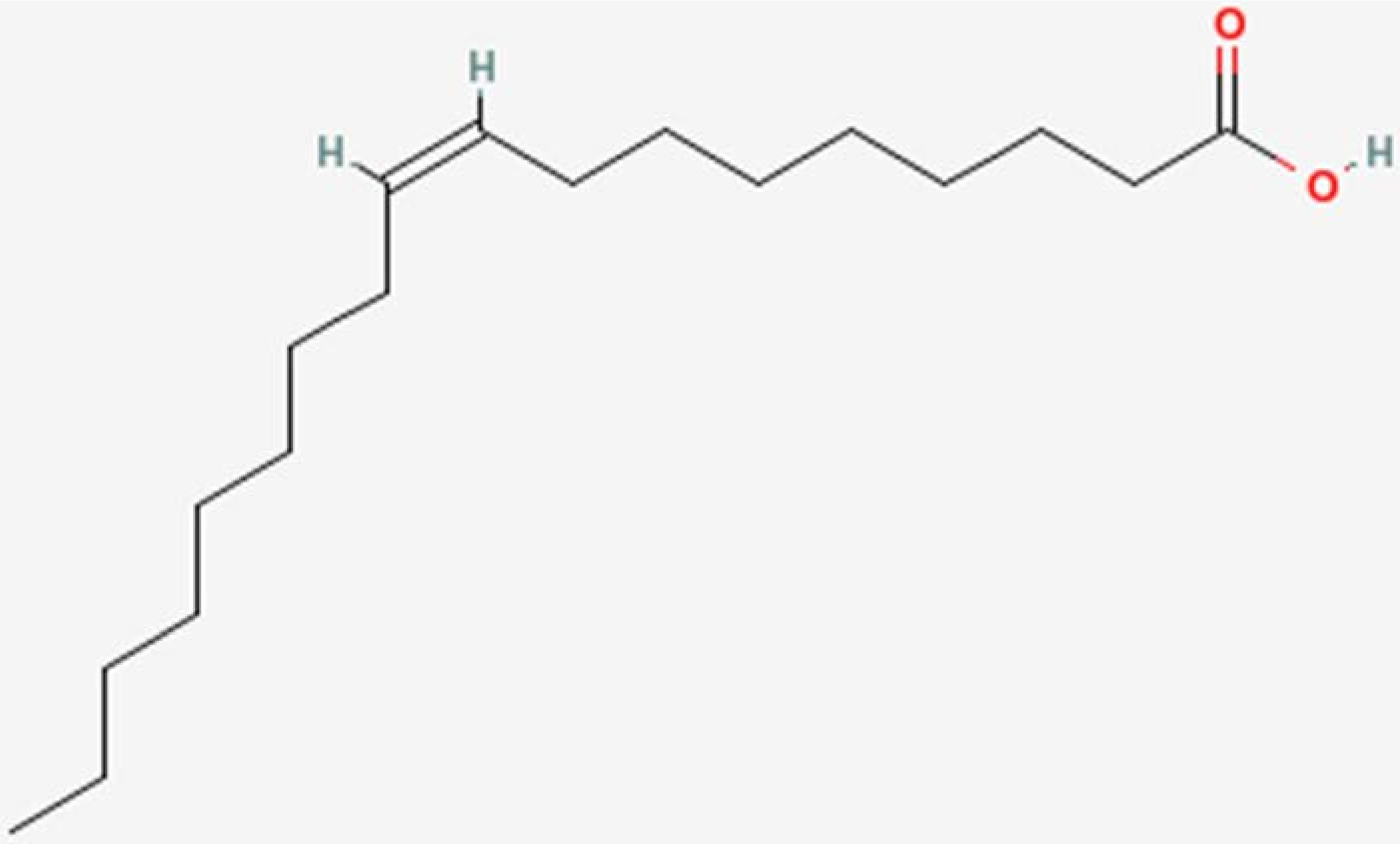
$C_{18}H_{36}O_2$

Stearic Acid

硬脂酸

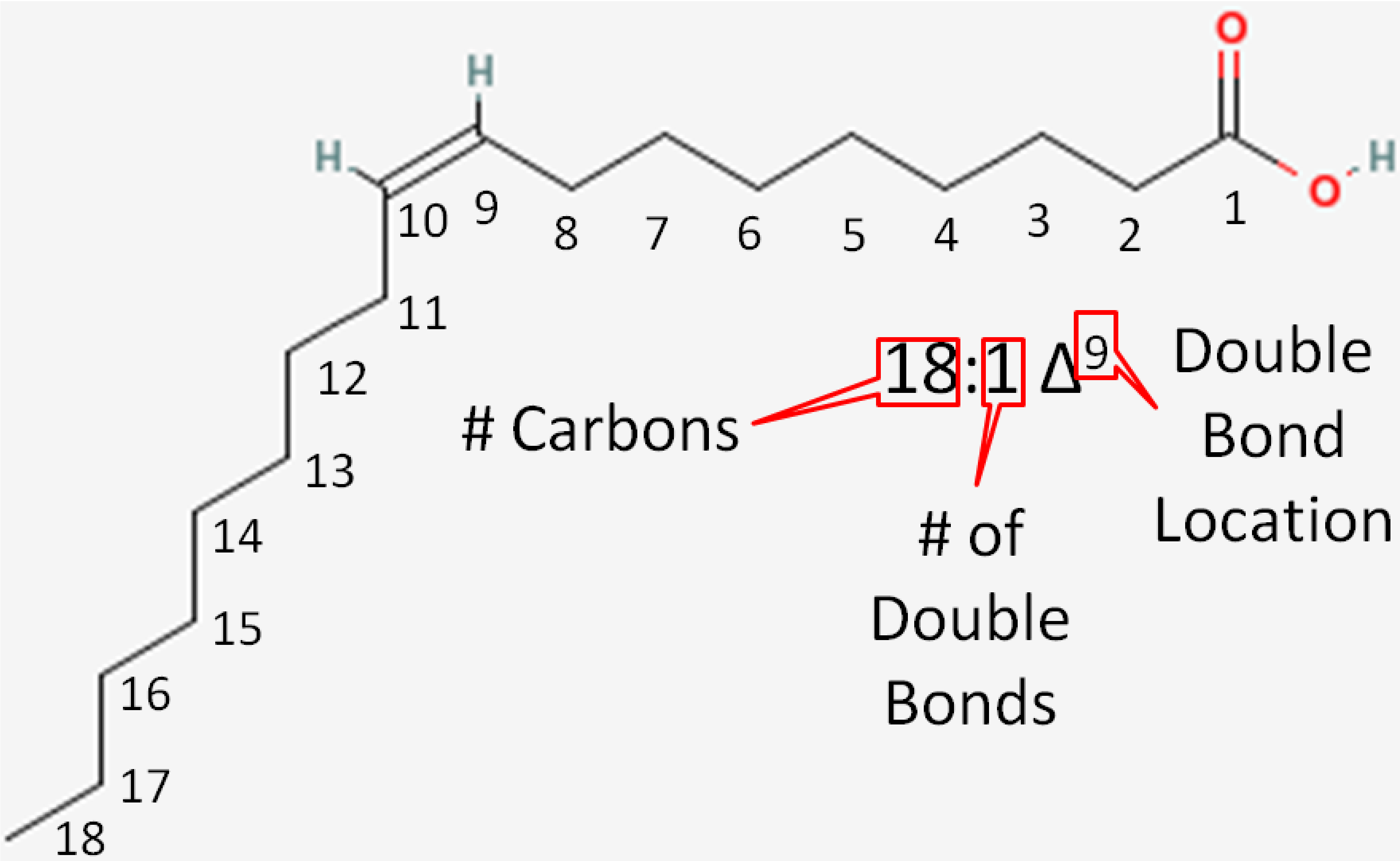
9-Octadecenoic acid

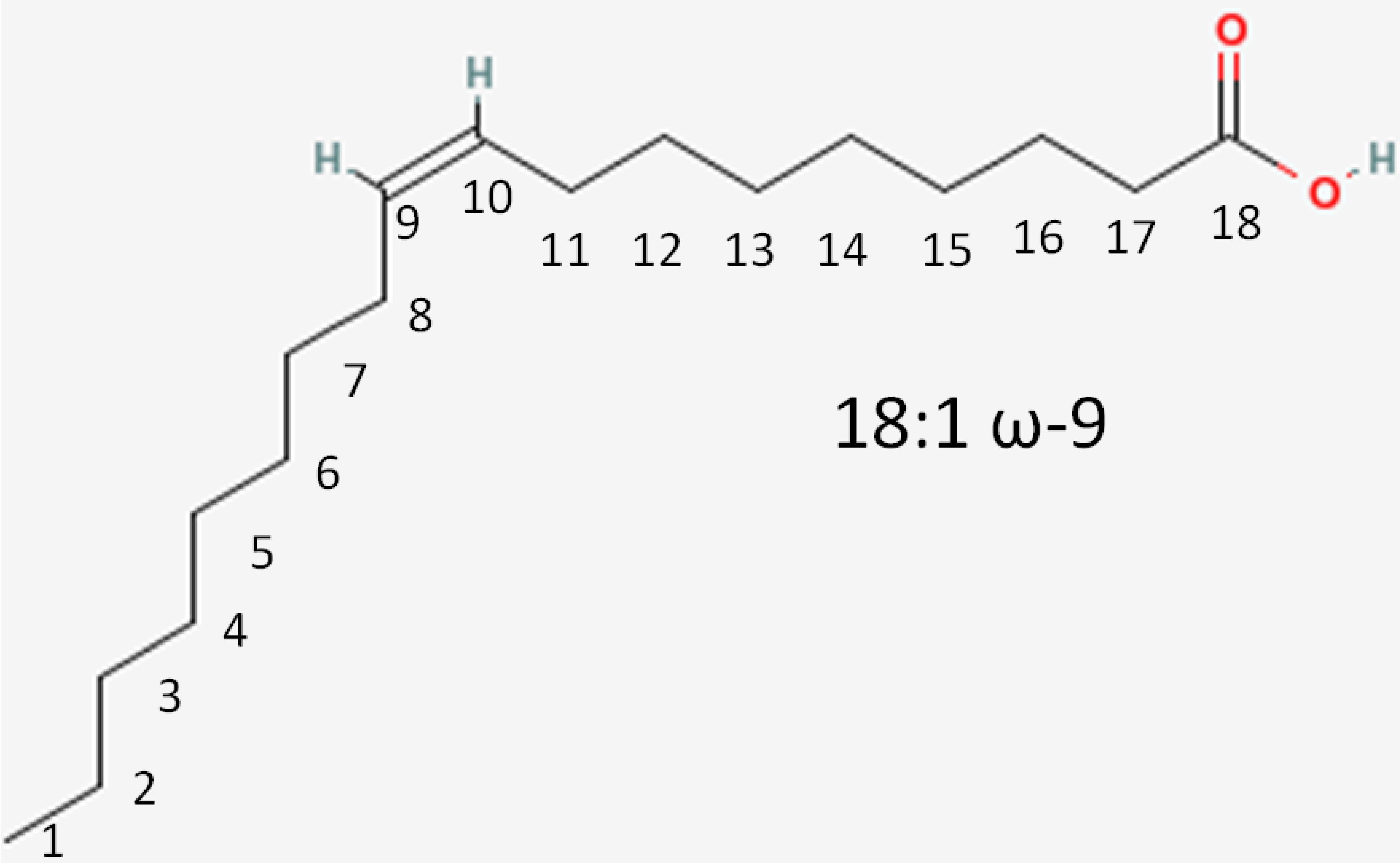
Oleic acid



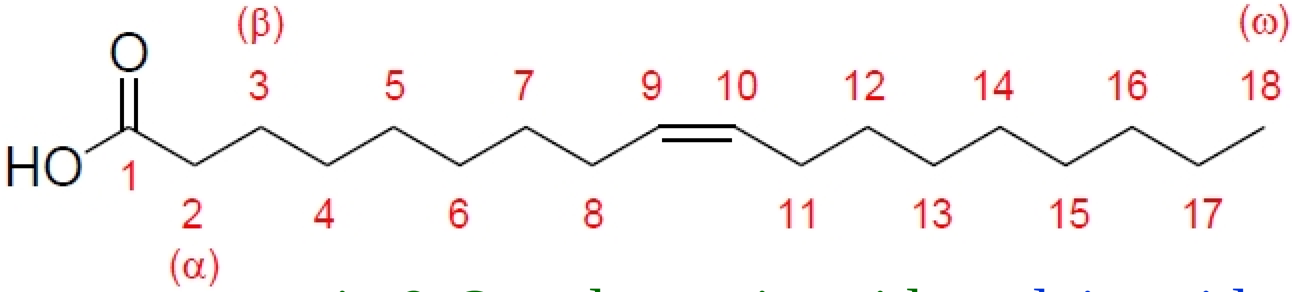
$C_{18}H_{34}O_2$

油酸


H
H
O
O
H
10
9
8
7
6
5
4
3
2
1
11
12
13
14
15
16
17
18
18:1 Δ9
Carbons
of Double Bonds
Double Bond Location


H
H
O
O
H
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
18:1 ω-9

(a)



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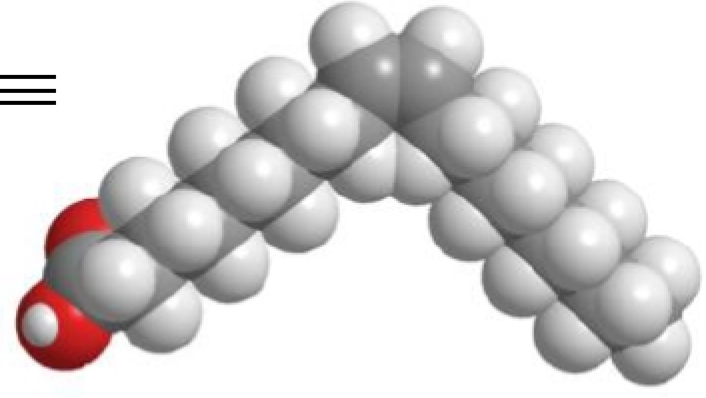
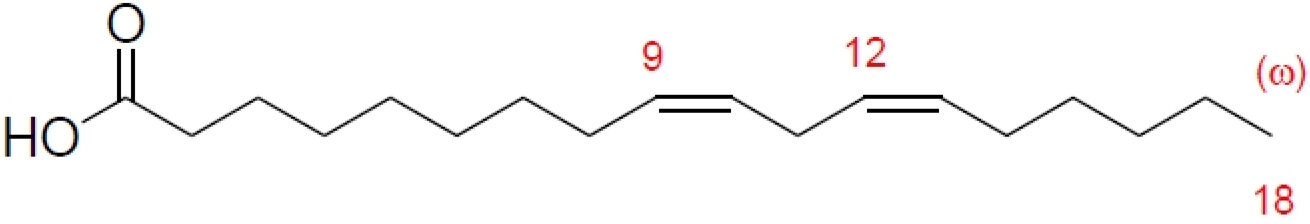


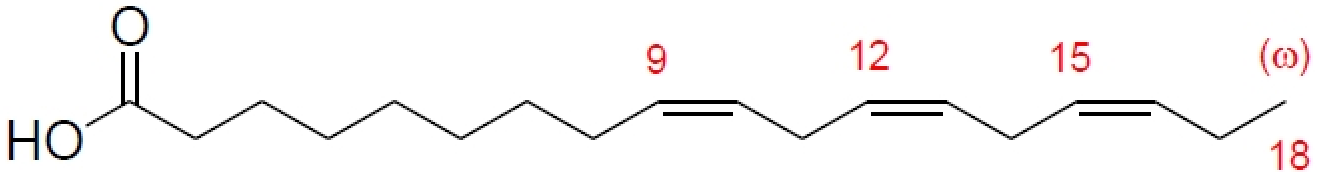
cis-9-Octadecenoic acid oleic acid

(b)



(9Z,12Z)-octadeca-9,12-dienoic acid linoleic acid

(c)



(9Z,12Z,15Z)-octadeca-9,12,15-trienoic acid linolenic acid

What you need to know

1. Sigma bond and pi bond
2. Alkane isomers
3. How to name an alkane